

Part VI — PMA parts and DER repairs

Descriptive reference. Sources are cited inline and listed at the end. Takes no view.

Contents

PMA parts and DER repairs.....	3
Translation-table rows answered.....	4
37. Why PMA exists, and the four approval bases	4
37.1 The gate: who may make a new part	4
37.2 What a Parts Manufacturer Approval is	6
37.3 The four approval bases.....	7
37.4 The production side	10
38. Critical parts and the hot-section problem	10
38.1 Two definitions that are easy to confuse	10
38.2 Why a turbine blade is the hard case	13
38.3 How long a PMA takes.....	14
39. EASA and the bilateral.....	14
40. DER repairs.....	16
40.1 The designee system.....	16
40.2 Manual repairs, major repairs and the form that approves them.....	16
40.3 Who develops and uses them, and how they save	19
41. The PMA market: HEICO, Chromalloy and others.....	20
41.1 HEICO Flight Support Group	20
41.2 Chromalloy	23
41.3 Other makers, and the manufacturer that once made PMA parts	25
41.4 Market size and penetration.....	26
42. Acceptance: airlines, lessors, and the OEM's responses	27
42.1 Airlines	27
42.2 Lessors	29
42.3 The manufacturer's responses.....	31
43. The FTAI–Chromalloy joint venture: what is filed and what is reported	34
43.1 Three levels of source.....	34
43.2 The approvals timeline	37
43.3 "Sold out through 2025" and the slot statements.....	38

43.4 What FTAI's management has said	38
44. Worked example: the parts bill with and without PMA.....	39
44.1 The setting and the inputs	40
44.2 The bill with the manufacturer's new parts, and the blade-price tension	41
44.3 Testing the claimed \$2 million against the arithmetic	43
Tensions carried in this Part	44
Unknowns carried in this Part	46
Sources	47

PART VI

PMA parts and DER repairs

This Part explains the two legal routes by which a company other than the engine's maker may put a new part into a CFM56, or repair one in a way the maker's manual does not describe. It covers the regulation, the hard case of hot-section parts, the European acceptance rules, the market that has grown up around approved alternative parts, how airlines, lessors and the manufacturer have responded, and what is filed and what is only reported about FTAI's own joint venture. A reader who finishes it will be able to read a parts-approval announcement, a lease's parts clause and a claimed per-visit saving with every term understood and every number placed against a total.

TRANSLATION-TABLE ROWS ANSWERED

This Part answers the rows of the Part III translation table that concern the price and supply of new material for an engine that is kept flying longer than planned. It answers (Part III §24, row: OEM parts escalation), because an approved alternative part is the only legal way to buy a new part for the CFM56 at a price the manufacturer does not set. It answers (Part III §24, row: hot-section material supply), because the high-pressure turbine blade, the part most often scrapped at a restoration, received its first approved alternative on 30 October 2025. It answers (Part III §24, row: engines kept past planned retirement), because an engine on its third or fourth shop visit needs new hot-section parts that an engine retired on schedule would never have bought. And it touches (Part III §24, row: lease extensions and return conditions), because whether a leased engine may carry an alternative part at all is decided in the lease. What it does not stress is the life-limited-part line: no approved alternative exists for any CFM56 life-limited part, so that line stays single-source at list price (Part I §4). The orchestrator fills the row identifiers.

37. Why PMA exists, and the four approval bases

Every new part that goes into a CFM56 must be made under an approval that a regulator granted to a named company. For most of the engine's life only one company held such an approval for most of its parts: the company that designed it. This section explains why that is so, what it does to prices, and the four ways a second company can obtain an approval of its own.

37.1 THE GATE: WHO MAY MAKE A NEW PART

An aircraft engine is certified as a design, and the company that owns the design is the only one that can make parts to it without a further approval. Part I §9 gave the short form of the two certificates involved; this Part owns them and gives the full form.

TYPE CERTIFICATE (TC) AND TYPE-CERTIFICATE HOLDER (TCH)

A type certificate is the Federal Aviation Administration's (FAA) approval of the design of an aircraft, an engine or a propeller, issued after the design has been shown to meet the applicable airworthiness standards. Its holder, the type-certificate holder, is the design's owner and the party responsible for keeping the design airworthy in service. For the CFM56 the holder is CFM International, the 50/50 joint venture of GE Aerospace and Safran Aircraft Engines (Part I §2). The holder's own new spare parts are produced under its production certificate, so for the holder the type certificate is also the gate to the spare-parts business. [eCFR, 14 CFR Part 21; FAA PMA design-approval page]

PRODUCTION CERTIFICATE (PC) AND PRODUCTION APPROVAL

A production certificate is the FAA's approval of a manufacturer's quality system, under which that manufacturer may produce parts to an approved design and release them as airworthy. The regulations treat the type-certificate holder's production certificate, a Parts Manufacturer Approval and a Technical Standard Order authorization (an approval for standardised articles such as instruments and seat belts) as the three forms of production approval. A replacement or modification part for a certificated product may be installed only if it was produced under one of them, or falls into a narrow exception: standard parts made to a published industry or military specification (the AN/MS hardware of the rules, meaning fasteners and fittings made to Air Force–Navy and Military Standard drawings), parts an owner or operator produces for its own aircraft, and parts a repair station makes in the course of a repair. [eCFR, 14 CFR Part 21 Subpart K; FAA PMA design-approval page]

The practical result is a single source. The manufacturer's new spares come off its own production certificate. Anyone else who wants to sell a new part for the engine must first obtain a Parts Manufacturer Approval, which §37.2 defines. Until that approval exists, the manufacturer is the only legal seller of that part number new. [eCFR, 14 CFR Part 21 Subpart K]

Used material is the other legal source of a part, and it sits outside this debate. Used serviceable material (USM), the parts recovered from retired or dismantled engines and re-released, was made by the manufacturer under its own production certificate, so no further design approval is needed to install it (Part V §31). But it competes for the same slot in a shop visit's parts bill as a new alternative part, and Part V §34 and this Part's §44 should be read together for that reason. [D4 §2.1; D3 §2.1]

Why does the single source matter enough to have produced a regulation, an industry and an antitrust complaint? Because of what it does to prices. The manufacturer publishes a catalogue list price for every spare part and revises it upward on a schedule; Part IV §28 carries the full treatment and Part I §4.3 the life-limited-part series. The figures the research reached are these. New engine spare-parts prices had been rising around 10% a year in 2022 and 2023, with a double-digit increase on 1 November 2022 and a high-single-digit increase in August 2023, and Safran's chief executive described the rule as pricing "above inflation by 3 to 4 points" for "at least for some years". [Aviation Week, late 2023] A full CFM56-7B life-limited-part set compounded at about 7% a year from 2008 to 2025 on the trade-press series Part I assembled (derived in Part I §4.3). [Part I, worked example 1.3] One independent shop reported that -7B parts rose 20–30% in 2023 after the August catalogue revision. [Magnetic Group via Aviation Week, 2024] Those readings cover different parts and years and the primer carries all of them. [reconciliation R-018]

The escalation lands on a bill that is mostly material. Material is 60–70% of a shop visit, labour 20–30%, repairs and other shop work 10–20%, according to one trade-press breakdown. [Air Cargo Week, 2025/26] And the CFM56 aftermarket is mostly transactional: Safran describes about

85% of its CFM56 aftermarket revenue as spare parts and time-and-materials work rather than long-term contracts. [Safran via Aviation Week, c. 2023/24] So on this engine the manufacturer's main capture is the parts line, whoever runs the shop, and a shop that is not the manufacturer has only three ways to reduce that line: used parts instead of new (Part V), repair instead of replacement (§40), and an approved alternative part instead of the manufacturer's (§37.2 onward). [D2 §2.2]

WORKED EXAMPLE 6.1 — WHAT CATALOGUE ESCALATION DOES TO ONE BLADE'S PRICE, AND TO THE SAVING FROM AN ALTERNATIVE

Inputs: a new manufacturer's CFM56 high-pressure turbine blade at about \$20,000 (sourced: In Practise expert interview, an interview digest, undated; the alternative listing of about €35,000 is in §44). Escalation of about 10% a year (sourced: Aviation Week, late 2023, "around 10% per year" for new spares) or 7% a year (derived: Part I §4.3 on the life-limited-part series). Five years (assumed). A set of 80 blades (a recollection, unverified in the research, D4 U16).

at 10% a year: $20,000 \times 1.10^5 = 32,210$ per blade; set of 80 = 2,576,800
 at 7% a year: $20,000 \times 1.07^5 = 28,051$ per blade; set of 80 = 2,244,100
 starting set of 80: $20,000 \times 80 = 1,600,000$

So a part priced at \$20,000 today is priced at roughly \$28,000 to \$32,000 in five years on either escalation reading, and a set of them moves from \$1.6M to \$2.2–2.6M. If an alternative part sells at a fixed discount d to the manufacturer's price, the dollar saving per set grows with the manufacturer's price: at $d = 30\%$ it is \$480,000 today and \$670,000 to \$770,000 in five years. Limits of the example: the per-blade price comes from an interview, not a catalogue; the blade count is unverified; whether any alternative maker prices as a fixed discount is not disclosed (D4 U2); and §44 shows that the same blade line, priced from a consultancy's cost shares instead of per blade, comes out several times smaller (R-015). The point here is the mechanism: escalation on a single-source part is what gives a second source its market. [In Practise, undated; Aviation Week, late 2023; D4 §2.8]

37.2 WHAT A PARTS MANUFACTURER APPROVAL IS

Part I §5 gave a short definition because the high-pressure turbine blade needed one there. This is the full form.

PARTS MANUFACTURER APPROVAL (PMA)

A Parts Manufacturer Approval is the FAA's combined design-and-production approval, issued under Title 14 of the Code of Federal Regulations (CFR) Part 21 Subpart K, that lets a company other than the type-certificate holder legally make and sell a replacement or modification part for a type-certificated aircraft, engine or propeller. It is a design approval, in that the FAA agrees the part's design meets the airworthiness standards that apply to the product it goes on, and a production approval, in that the FAA agrees the holder's quality system can make every unit to that design. It is issued as a letter with a supplement listing each approved part number and the models it is eligible for; the parts are marked "FAA-PMA" with the maker's name and part number, and each shipment is released on FAA Form 8130-3. There are four approval bases (§37.3). PMA has historically concentrated in consumables and expendables; a cooled, coated turbine blade approved in October 2025 is treated in the trade as the hot-section milestone (§38). No PMA exists for any CFM56 life-limited part. [eCFR, 14 CFR Part 21 Subpart K and 14 CFR 45.15; FAA Order 8110.42D; FAA PMA design-approval page; D4 U7]

14 CFR PART 21 SUBPART K

Part 21 Subpart K is the section of the FAA's certification regulations under which Parts Manufacturer Approvals are issued. Section 21.303 sets the design showing: the applicant must provide "test reports and computations necessary to show that the design of the article meets airworthiness requirements, unless the applicant shows that the design is identical to an article covered under a type certificate", and where the design came by a licensing agreement, evidence of that agreement. Section 21.307 requires a quality system meeting the standard of section 21.137 (the same standard a production-certificate holder meets), and section 21.316 lists the holder's continuing obligations once approved. [eCFR, 14 CFR Part 21 Subpart K]

Two FAA offices divide the work. The design side is handled by an Aircraft Certification Office (ACO), or by engineers the FAA has designated to act for it (§40 describes the designee system). The production side is handled by a Manufacturing Inspection District Office (MIDO), which inspects the applicant's quality system and the conformity of the test articles. Both must be satisfied before the supplement is issued. [FAA Order 8110.42D; FAA PMA design-approval page] The release tag that accompanies each shipment is FAA Form 8130-3, which Part I §9 defined: for a new part it certifies that the part was produced under an FAA production approval and conforms to its approved design. A part not produced under a production approval cannot carry one. [Aviation Suppliers Association, 2016, via Part I §9]

37.3 THE FOUR APPROVAL BASES

Section 21.303 draws one line: either the applicant proves the design by its own tests and calculations, or it shows the design is identical to the manufacturer's. FAA guidance turns that

line into four approval bases, which differ in what the applicant has to possess and what it has to prove. [FAA AC 21.303-3; FAA AC 21.303-4; FAA Order 8110.42D]

TEST AND COMPUTATION (T&C)

Test and computation is the PMA basis an applicant uses when it has none of the manufacturer's design data. It reverse-engineers the part: it measures examples of the manufacturer's part, determines the material and any coatings, and writes its own drawings and process specifications. It then proves by its own analysis and testing that the part meets the airworthiness standards that apply to the product (for an engine part, 14 CFR Part 33) and does not degrade the engine's own compliance. It is the only basis on which a company can compete with the manufacturer on a part the manufacturer has not licensed, and Chromalloy's CFM56 high-pressure turbine blade is a test-and-computation part. [FAA AC 21.303-3; FAA AC 33-8; D4 §2.2]

IDENTICALITY (WITHOUT A LICENSING AGREEMENT; BY LICENSING AGREEMENT)

Identity is the PMA basis in which the applicant shows that its design is identical to the type-certificate holder's in every respect that matters. Without a licence, the applicant usually holds the manufacturer's drawings legitimately, for example as a former supplier; FAA Notice 8110.55 of 1996 set the policy. With a licence, the type- or supplemental-type-certificate holder has licensed its design data and the licence itself is the evidence; this is how a manufacturer's subcontractor, or a company the manufacturer has chosen to work with, obtains a PMA. For a critical or life-limited part approved by identity the FAA additionally requires evidence that the manufacturing, inspection and test processes are identical, not only the geometry (§38). [FAA Notice N 8110.55, 1996; FAA Order 8110.42D; FAA AC 21.303-4]

SUPPLEMENTAL TYPE CERTIFICATE (STC)

A Supplemental Type Certificate is the FAA's approval of a change to a type-certificated design. As a PMA basis, the part is the subject of its own design approval, the STC, and the PMA becomes the production approval for that STC's parts. The STC route matters for international acceptance: under the FAA–EASA bilateral, a critical part approved by test and computation is accepted in Europe only through an STC that EASA has validated (§39). The same instrument is the approved modification design a conversion house owns for a passenger-to-freighter conversion, which Part II §15 glossed. [FAA AC 21.303-4; FAA–EASA TIP Rev 7; D4 §2.2, §2.5]

EXHIBIT 6.1 — THE FOUR PMA APPROVAL BASES: WHAT THE APPLICANT HOLDS, WHAT IT MUST SHOW, AND WHO USES EACH

Basis	What the applicant holds	What it must show the FAA	Typical user	CFM56 example in the research
Test and computation	No manufacturer data; its own reverse-engineered drawings and process specifications	Test reports and computations proving the design meets the applicable standards (14 CFR Part 33 for an engine part) without degrading the engine's compliance; for turbine parts, the showings of AC 33-8	An independent competing on an unlicensed part	Chromalloy CFM56-5B/7B HPT blade, approved 30 Oct 2025
Identity without a licence	The manufacturer's drawings held legitimately (e.g. a former supplier)	That the design is identical; for a critical or life-limited part, that processes and inspections are identical too	A former or current supplier acting on its own account	None found
Identity by licence	A licence from the TC or STC holder to its design data	Evidence of the licence, which establishes that the data are FAA-approved and identical	A manufacturer's subcontractor or chosen partner	None found
Supplemental Type Certificate	Its own approved design change	The STC's own showing; the PMA is then the production approval for the STC's parts	Any applicant needing a separate design approval, including for EASA acceptance of a critical part	None found

Source: eCFR, 14 CFR 21.303; FAA Advisory Circulars 21.303-3 and 21.303-4; FAA Order 8110.42D; FAA AC 33-8; FAA Notice N 8110.55 (19 July 1996); Chromalloy release, 30 Oct 2025. Compiled in D4 §2.2. Accessed 2026-10-03.

Read across the rows and the pattern is that the harder the data are to obtain, the more the applicant must prove. A licensee proves almost nothing beyond the licence. A former supplier proves identity. An independent with no data proves airworthiness from scratch. That is why the independent PMA industry grew up in parts whose airworthiness is easy to prove, and why §38 is a separate section. [D4 §2.2–2.3]

WHY THE FOURTH BASIS MATTERS ABROAD

Inside the United States a PMA part approved by any of the four bases is installable on any eligible engine. The bases matter abroad. The European acceptance rules (§39) accept an FAA PMA directly only in defined cases, and for a part the manufacturer designated as critical the only direct route is a Supplemental Type Certificate that the European agency has itself validated. So an applicant's choice of basis is partly a choice of addressable fleet: a test-and-computation approval alone may serve US-registered engines, while serving EU-registered engines may require the fourth route as well. Whether that applies to the CFM56 turbine blade and vane is one of this Part's open questions. [FAA–EASA TIP Rev 7; D4 §2.5, U5]

37.4 THE PRODUCTION SIDE

An approved design is half of a PMA. The other half is an approved factory. The applicant's quality system must meet 14 CFR 21.137, the same standard applied to the type-certificate holder's own production certificate. Before issuing the supplement the FAA checks the conformity of the test articles to the drawings, reviews the test and analysis reports, and inspects the quality system. After issue, section 21.316 obliges the holder to keep the system in place, to report failures and defects, and to allow FAA inspection. [eCFR, 14 CFR 21.307 and 21.316; FAA Order 8110.42D] Every unit then leaves on an 8130-3 that references the PMA. The approval is therefore to a company and a part number, not to a design in the abstract: a second company wanting to make the same part must obtain its own PMA, and a PMA holder that changes its process must show the change does not alter conformity. [FAA Order 8110.42D; D4 §2.2]

38. Critical parts and the hot-section problem

Alternative parts were for decades a business in seals, bushings, fasteners, filters, brackets and non-rotating structure. The parts that cost the most in a CFM56 shop visit are not those. This section explains the two classifications that govern what a maker must prove, why a turbine blade sits at the hard end of both, and what the research found about how long the hard case takes.

38.1 TWO DEFINITIONS THAT ARE EASY TO CONFUSE

Two different categories are in play, and they come from two different documents.

The first is the life-limited part, which Part I §4 defined: a rotating or pressure-bearing part whose failure the engine case cannot contain, certified for a fixed number of flight cycles and scrapped at that limit regardless of condition. On the -7B the limits are 20,000 cycles in the high-pressure compressor and turbine, 25,000 in the low-pressure turbine and 30,000 in the fan and

booster. [Part I §4.1; AJW and StandardAero mini-packs] The list of such parts, and each one's limit, sits in a specific section of the engine's certification data.

AIRWORTHINESS LIMITATIONS SECTION (ALS) AND INSTRUCTIONS FOR CONTINUED AIRWORTHINESS (ICA)

The Instructions for Continued Airworthiness are the manufacturer's approved maintenance instructions for a type-certificated product, attached to the type certificate: the Engine Shop Manual and the component maintenance manuals (Part I §9) are part of them. The Airworthiness Limitations Section is the part of those instructions that lists the mandatory life limits and inspection intervals, which no operator may exceed; every life-limited part's cycle limit appears there. The two matter in this Part because a part's appearance in the Airworthiness Limitations Section is the usual trigger for the "critical" designation below, and because a repair that departs from the Instructions for Continued Airworthiness needs separate approval (§40). These definitions are drawn from first principles and from the regulatory structure the dossier cites; the research did not quote the regulation's own text.

[D4 §2.3, §2.6; registry, Part VI row]

The second category comes from the agreement between the FAA and its European counterpart, and it is the one that decides how an approval travels.

CRITICAL COMPONENT (CRITICAL PART)

Under the FAA–EASA bilateral's Technical Implementation Procedures, a critical component is "a part identified as critical by the design approval holder during the product certification process or otherwise by the Authority for the State of Design. Typically, such components include parts for which a replacement time, inspection interval, or related procedure is specified in the Airworthiness Limitations section or certification maintenance requirements." Every life-limited part is therefore a critical component. Whether a given turbine blade or vane is one depends on how the type-certificate holder designated it during certification, and that designation is not public in anything the research retrieved. The classification matters because it selects which European acceptance clause applies (§39) and because, for an identity approval, it adds the requirement that processes and inspections be identical, not only geometry. [FAA–EASA TIP Rev 7, as returned by search; FAA Order 8110.42D; D4 §2.3, U5]

EXHIBIT 6.2 — THREE CLASSES OF CFM56 PART AND WHAT EACH MEANS FOR AN ALTERNATIVE MAKER

Class	Defining document	CFM56 examples	Replaced when	What a PMA applicant must additionally show	PMA found in the research
Life-limited part	Airworthiness Limitations Section of the engine's certification data	Fan disk, booster spool, HPC spools and disk, HPT disk, LPT disks, shafts, certain seals	At the certified cycle limit, whatever its condition	Critical by definition under the bilateral; identity requires identical processes; T&C requires a full life substantiation	None for any CFM56 life-limited part (D4 U7)
Critical component that is not life-limited	TCH designation during certification; typically carries an inspection interval in the Airworthiness Limitations Section	Not public for the CFM56 in the sources retrieved	On condition or at an interval	Same process-identity requirement; EASA acceptance only via a validated STC if approved by T&C	Unknown whether the HPT blade and vane are so designated (U5)
On-condition hot-section part	Engine Shop Manual limits	HPT blades, HPT stage-1 vanes (the HPT nozzle), combustor liners, LPT blades and vanes, shroud segments	When inspection finds it beyond repair limits	For T&C: material, cooling, durability and safety-analysis substantiation per AC 33-8	HPT blade (30 Oct 2025); HPT vane 1; LPT vane; HPT shroud per one reading (§41)

Source: FAA–EASA TIP Rev 7 (definition of critical component, as returned by search); FAA Order 8110.42D; FAA AC 33-8; Part I §4 (life-limited parts and limits); Chromalloy releases, Oct–Nov 2025; D4 §2.3 and U5, U7. Accessed 2026-10-03.

The third row is where the money in a restoration sits. The high-pressure turbine blades and vanes are not life-limited; they are replaced when inspection finds them beyond the manual's repair limits, and that happens at the hot end of the engine faster than anywhere else. Part I §5 explained that most of the lost temperature margin is recovered in that module, and one consultancy puts the blades and vanes at 15–25% of a shop visit's parts cost. [Safe Fly Aviation, 2026,

consultancy blog; Part I §5] So the on-condition hot-section row, not the life-limited row, is where an approved alternative changes a bill.

38.2 WHY A TURBINE BLADE IS THE HARD CASE

Two regulatory facts make the hot section hard for an applicant with no manufacturer data.

The first is the showing itself. For a test-and-computation approval of a turbine engine part the FAA's product-specific guidance is Advisory Circular (AC) 33-8. Under it the applicant must substantiate the part's material properties, its cooling and thermal behaviour, its durability, and its effect on the engine's safety analysis by its own testing, because it has no access to the manufacturer's data on any of them. [FAA AC 33-8; D4 §2.3] A high-pressure turbine blade is a hollow casting with internal cooling passages, made from a single-crystal nickel alloy and coated to survive gas hotter than its own melting point. This primer names those three facts and stops, since the depth that matters here is what they do to the approval: each is a property the applicant must measure, reproduce and prove, with no reference data from the designer.

The second is the identity rule for critical and life-limited parts. If the basis is identity, the FAA's procedures require the applicant to show that the manufacturing processes (casting, forging, heat treatment, coating) and the inspection and test procedures produce a part identical to the original, not merely that the drawing matches. [FAA Order 8110.42D, as cited in D4 §2.3] For a part whose life depends on its microstructure, that is a showing few companies outside the manufacturer's own supply chain can make.

WHY A BLADE IS HARDER THAN A BRACKET

A bracket's airworthiness is a matter of geometry, material and static strength, all of which an applicant can measure from an example and prove by a calculation and a pull test. A cooled turbine blade's airworthiness is a matter of how a particular alloy, grown as a single crystal and coated, behaves over thousands of hours in gas hotter than its melting point, under centrifugal load, with cooling air flowing through passages whose shape the applicant must infer. The regulation does not forbid an independent from proving that; AC 33-8 tells it how. But the proof is an engine-level test and durability programme the applicant funds itself, and the trade treats the first such approval for the CFM56-5B/7B, announced on 30 October 2025, as a milestone for exactly that reason: the applicant called its blade "the only aftermarket alternative to the OEM manufactured new part" for a platform with "over 20,000 engines" in service. [FAA AC 33-8; Chromalloy release, 30 Oct 2025; D4 §2.3]

Those two facts explain the historical shape of the industry. Approved alternatives concentrated in consumables and expendables, meaning the seals, bushings, fasteners, filters, brackets and non-rotating structural pieces whose airworthiness can be proven with modest test programmes. [D4 §2.3] They did not reach life-limited rotating parts at all: the research found no PMA for any CFM56 disk, spool or shaft, and that line of the parts bill remains single-source. [D4 U7; Part I §4.4]

And they reached the cooled hot-section airfoils only recently, and only after a programme whose length §38.3 sets out as far as the sources allow.

38.3 HOW LONG A PMA TAKES

Neither the FAA nor the companies disclose the interval from application to approval in anything the research retrieved, so this is an open question (D4 U4). Two datapoints bracket it. At the simple end, HEICO's annual report says the company has been adding new approvals at a rate of "approximately 300 to 500" a year by one dossier's reading of the filing, or "approximately 400 to 550" by another's, across a portfolio dominated by less complex parts; simple approvals are evidently a matter of months, not years. [HEICO 10-K FY2025, two readings; reconciliation R-024] At the hard end, third-party commentary dates the FTAI–Chromalloy partnership to 2018 and the FAA approval of the CFM56 turbine blade came on 30 October 2025, about seven years later; the application date is not public, and the 2018 date is itself an investor-blog figure that §43 sets beside what is filed. [Komodo Capital and Kairos Research, investor blogs, undated; Chromalloy release, 30 Oct 2025; D4 §2.4] The FAA's own registry of approvals would carry the dates, and the research did not read it. [D4 U4]

39. EASA and the bilateral

An FAA approval is a US approval. Roughly half the CFM56 fleet is registered outside the United States, and for the largest single bloc of it, the European Union, acceptance of US approvals is governed by a treaty and the procedures written under it. This section sets out those procedures as the research found them and names the question they leave open for the CFM56 hot-section parts.

TECHNICAL IMPLEMENTATION PROCEDURES (TIP)

The Technical Implementation Procedures are the document by which the FAA and the European Union Aviation Safety Agency (EASA) implement the US–EU Bilateral Aviation Safety Agreement for airworthiness and environmental certification; the current edition is Revision 7 with Amendment 1. Among much else, they set the rules under which EASA accepts FAA Parts Manufacturer Approvals for installation on EASA-certificated or EASA-validated products. As returned by search, EASA accepts an FAA PMA directly, without further showing, when the part is not a critical component and was approved by test and computation or by identity without a licence; or when it conforms to design data licensed from a type- or supplemental-type-certificate holder whose certificate EASA has validated; or when it is a critical component approved through an FAA supplemental type certificate that EASA has validated. [FAA–EASA TIP Rev 7 with Amendment 1; TIP Rev 6 (PDF); D4 §2.5]

A caution on the wording. The dossier recorded that a search summary rendered the first clause as "identity without a licensing agreement" and advised that the exact text of Revision 7 be checked before it is quoted. The research did not read the document itself, so this Part reports the three clauses as the dossier recorded them and does not quote them. [D4 §2.5]

EXHIBIT 6.3 — THE THREE TIP ACCEPTANCE CLAUSES FOR AN FAA PMA PART ON AN EU-REGISTERED ENGINE, AS THE RESEARCH RECORDED THEM

Clause	Condition on the part	Condition on the approval basis	What accompanies the part	Where the CFM56 HPT blade and vane fall
1	Not a critical component	Approved by test and computation, or by identity without a licensing agreement	FAA Form 8130-3 with the EASA-acceptance statement; no further showing	Here, if the TCH did not designate them critical (unknown, D4 U5)
2	Any	Conforms to design data obtained under a licence from a TC or STC holder, where EASA has validated that TC or STC	8130-3; no further showing	Not applicable: the blade is a T&C part, not a licensed copy
3	A critical component	Approved through an FAA STC that EASA has validated	8130-3 referencing the validated STC	Here, if the TCH designated them critical and no validated STC exists (unknown, U5)

Source: FAA–EASA Technical Implementation Procedures Rev 7 with Amendment 1 (web) and Rev 6 with Amendments 1 and 2 (PDF), as returned by search and recorded in D4 §2.5; the exact clause wording was not read. FAA EASA FAQ (PDF) for the Form 1 equivalence. Accessed 2026-10-03.

The practical consequence is that a non-critical test-and-computation part travels to Europe on its FAA release tag, while a critical test-and-computation part needs the supplemental-type-certificate route plus EASA's validation of that certificate. [D4 §2.5] Part I §9 defined EASA Form 1 as the European release certificate, the equivalent of the 8130-3, issued by a holder of a European production organisation approval; a US-made PMA part moving into the EU system under the clauses above does so on its 8130-3, not on a Form 1, since its maker holds an FAA approval rather than a European one. [FAA EASA FAQ; Part I §9]

The question the research could not close is the one that decides the addressable fleet. Whether CFM International designated the CFM56-5B/7B high-pressure turbine blade and stage-1 vane as critical components during certification is not public in the sources retrieved; whether Chromalloy has obtained EASA acceptance for them by any clause was likewise not found. [D4

U5] If the parts are not critical, clause 1 applies and the FAA approval of October 2025 serves EU-registered engines directly. If they are critical, clause 3 applies and a validated supplemental type certificate is required, which the research did not find. This Part presents both branches and does not choose between them. Acceptance by other authorities, including the UK Civil Aviation Authority (CAA), Transport Canada Civil Aviation (TCCA), Brazil's ANAC (its national civil aviation agency), the Civil Aviation Administration of China (CAAC) and India's Directorate General of Civil Aviation (DGCA), runs under separate bilaterals and national rules, and none was retrieved. [D4 U6]

40. DER repairs

A new part, whoever makes it, is one way to restore an engine. Repairing the part already in it is the other, and the manufacturer's manual does not describe every repair that is possible. This section explains who may approve a repair the manual does not cover, on what document, how such a repair differs from one taken from the manual, who develops and uses them, and how they save money. Part I §9 gave the short form of the designee and the repair; this Part owns both.

40.1 THE DESIGNEE SYSTEM

The FAA does not have enough engineers to examine every set of engineering data itself. It appoints private engineers to do so on its behalf within defined limits.

DESIGNATED ENGINEERING REPRESENTATIVE (DER)

A Designated Engineering Representative is an individual engineer appointed by the FAA under 14 CFR Part 183 to examine engineering data and make findings of compliance on the FAA's behalf within a delegated technical area, such as structures, powerplant, engines or systems. A company DER is employed by a manufacturer or repair station and may act only on that company's data; a consultant DER may act for any client. To approve the data for a major repair the designee needs a specific authorisation, "Special – Major Repairs", from the managing Aircraft Certification Office, granted one-time or as part of the designee's standing delegation. The governing document is FAA Order 8110.37, the DER Handbook, currently revision F. [FAA Order 8110.37F; D4 §2.6]

40.2 MANUAL REPAIRS, MAJOR REPAIRS AND THE FORM THAT APPROVES THEM

The manufacturer publishes its approved repairs in the Engine Shop Manual and the component maintenance manuals, which Part I §9 defined and which form part of the Instructions for Continued Airworthiness attached to the type certificate (§38.1). A shop that follows the manual needs no further approval: the manual is already FAA-approved data. [D4 §2.6] The question arises when a shop wants to do something the manual does not describe.

MAJOR REPAIR

A major repair is a repair the manufacturer's manual does not cover, or a repair of a part the manual says must be scrapped, that does not qualify as minor. Major repairs "must be accomplished in accordance with technical data approved by the Administrator", meaning the FAA or someone acting under its delegation. The approval covers the data: the repair scheme, the substantiating analysis, the process specifications and any test. The repair itself is then performed by a certificated repair station under its own certificate. [FAA Order 8110.37F; FAA Order 8300.16; D4 §2.6]

FAA FORM 8110-3 (STATEMENT OF COMPLIANCE WITH AIRWORTHINESS STANDARDS)

Form 8110-3 is the document on which a Designated Engineering Representative records an approval, or a recommendation for approval, of engineering data, including the data for a repair or alteration. The designee keeps the original and sends copies to the FAA's managing office and to the owner, operator or repair station that requested the approval. The form approves data, not a part: the repair station that performs the repair releases the repaired part on FAA Form 8130-3 (Part I §9), and that release references the 8110-3 under which the repair data were approved. The FAA notes that designee-approved data "may not be adequate to cover every aspect of the repair", which is why the repair station's own certificate and release remain necessary. [FAA Order 8110.37F; FAA, Documenting Compliance Findings (Form 8110-3 guidance); D4 §2.6]

DER REPAIR

A DER repair is a major repair whose data a Designated Engineering Representative approved on FAA Form 8110-3, rather than being taken from the manufacturer's shop manual. The approved data may be one-time, covering a specific serial-numbered part, or multiple-use, repeatable on any eligible part; a multiple-use approval is what turns a repair into a product a shop can sell. The part remains the manufacturer's part; nothing is remanufactured. A DER repair saves money by restoring a part the manual would scrap, so that no new part of any origin is bought, or by restoring it more cheaply than the manual's repair. [FAA Order 8110.37F; D4 §2.6–2.7; D2 §2.2]

EXHIBIT 6.4 — MANUAL REPAIR, DER REPAIR AND PMA PART: WHO APPROVES WHAT, ON WHICH DOCUMENT, AND HOW EACH SAVES

Route	What it is	Source of the engineering data	Who approves the data	Document	What leaves the shop	How it saves	Acceptance questions
Manual repair	A repair described in the Engine Shop Manual or component manual	The type-certificate holder	The FAA, through its approval of the manual as part of the Instructions for Continued Airworthiness	The manual itself; part released on 8130-3	The manufacturer's part, repaired to the manufacturer's scheme	Repair instead of a new part	None: accepted everywhere the manual is
DER repair	A major repair outside the manual, or of a part the manual scraps	An independent engineering organisation, repair house, airline or MRO	An FAA designee with "Special – Major Repairs" authorisation	FAA Form 8110-3 (data); 8130-3 (release) referencing it	The manufacturer's part, repaired to an independently approved scheme	Repairs what the manual would scrap, or repairs more cheaply	Lease clauses may exclude it (§42); EU acceptance of the data not verified (D4 U8)
PMA part	A new part made by someone other than the type-certificate holder	The PMA holder (T&C or identity) or a licensor	The FAA (Aircraft Certification Office and Manufacturing Inspection District Office)	PMA letter and supplement; 8130-3 per shipment	A new, non-manufacturer part marked "FAA-PMA"	New part at a price the manufacturer does not set	Lease clauses may exclude it (§42); TIP clauses decide EU acceptance (§39)

Source: FAA Order 8110.37F (DER Handbook); FAA Order 8300.16; FAA Form 8110-3 guidance; eCFR 14 CFR Part 21 Subpart K; FAA–EASA TIP Rev 7; D4 §2.2, §2.5, §2.6 and §2.9. Accessed 2026-10-03.

How a DER repair differs from a manual repair comes down to three things, and the table shows them: the source of the data (independent engineering against the type-certificate holder), the approving authority (an FAA designee against the FAA's approval of the manufacturer's manual), and acceptance by third parties, meaning lessors and foreign regulators, which §42 and the open question on EU acceptance address. [D4 §2.6] The manufacturer's own 2018 commitments, described in §42.3, refer throughout to "non-CFM parts and/or repairs", treating PMA parts and

DER repairs as one category for warranty and service purposes. [IATA–CFM agreement, 2018; MARPA, 1 Aug 2018]

40.3 WHO DEVELOPS AND USES THEM, AND HOW THEY SAVE

Three kinds of organisation develop DER repairs. Independent turbine-part repair houses are the first: Chromalloy's historic core business is turbine-part repair and coating, and the trade press describes it as positioned in the "PMA and designated engineering repair sector". [FlightGlobal, Chromalloy profile, undated; D4 §2.6] Airlines with in-house engineering are the second: Southwest Airlines was reported as evaluating PMA and DER options for its CFM56-7B fleet as it phases out its 737-700s. [Aviation Week, c. 2023] Independent maintenance, repair and overhaul (MRO) shops are the third. [D4 §2.6]

The saving is on the repair line, and the size of that line is the first thing to establish. Repairs and other shop activities are 10–20% of a shop visit in the trade-press breakdown Part I carried; material is 60–70%. [Air Cargo Week, 2025/26] A repair that keeps a part in service avoids a purchase on the material line, so its saving is the new part's price less the repair's cost, and the research did not reach a repair price for any CFM56 hot-section part. What it did reach is the cadence that decides whether a repair is available at all. According to an expert interview, at a first shop visit, around 20,000 cycles, airlines typically repair the high-pressure turbine blades rather than replace them; at a second, around 30,000 cycles, the hot-section parts typically cannot be repaired again and a new part is needed. [In Practise, undated, interview digest] So a repair, manual or DER, is the lever at the first visit, and a new part, manufacturer's or PMA, is the lever at the second. The manufacturer has been working the repair lever too: CFM says it has industrialised about 400 CFM56 repairs since early 2023, meaning it has developed them into repeatable production processes, including automated laser welding of high-pressure turbine blades that cuts repair turnaround by about a third. [CFM International, "CFM boosts HPT blade output and repair", 2025/26, date not captured]

WORKED EXAMPLE 6.2 — SIZING THE REPAIR LINE AND THE NEW-PARTS LINE IN A RESTORATION FROM THE PUBLISHED SHARES

Inputs: a CFM56-7B performance restoration at \$1.2–1.6M (sourced: Safe Fly Aviation, consultancy blog, 2026, the range Part IV uses) or \$1.8–2.5M (sourced: Safe Fly Aviation, other page, 2026, the range Part I uses); both are carried under reconciliation rule E.1.4. Shares: material 60–70%, repairs and other shop work 10–20% (sourced: Air Cargo Week, trade press, 2025/26). HPT blades and vanes 15–25% of parts cost (sourced: Safe Fly Aviation, consultancy blog, 2026).

repair line, lower range: $0.10 \times 1,200,000 = 120,000$ to $0.20 \times 1,600,000 = 320,000$
 repair line, upper range: $0.10 \times 1,800,000 = 180,000$ to $0.20 \times 2,500,000 = 500,000$
 material line, lower range: $0.60 \times 1,200,000 = 720,000$ to $0.70 \times 1,600,000 = 1,120,000$
 material line, upper range: $0.60 \times 1,800,000 = 1,080,000$ to $0.70 \times 2,500,000 = 1,750,000$
 HPT blades and vanes as a share of material, lower range: $0.15 \times 720,000 = 108,000$ to $0.25 \times 1,120,000 = 280,000$
 HPT blades and vanes as a share of material, upper range: $0.15 \times 1,080,000 = 162,000$ to $0.25 \times 1,750,000 = 437,500$

So on the published shares the repair line of a restoration is \$0.12–0.50M and the material line \$0.72–1.75M, depending on which consultancy range is used, and the turbine blades and vanes within material are \$0.11–0.44M. A DER repair that keeps a blade set in service at a first visit saves against the material line; a PMA blade at a second visit reduces it. Limits of the example: both consultancy ranges are contested (R-013); the shares come from two secondary sources, neither of which names a shop-level data set (R-014); and §44 shows that the same blades and vanes, priced per blade from an interview, cost several times these figures (R-015). The example sizes the lines on one set of sources; it does not reconcile them. [Safe Fly Aviation, 2026; Air Cargo Week, 2025/26; reconciliation R-013, R-014, R-015]

One question the research could not answer belongs here. EASA's equivalent of a DER repair is a repair design approved by a holder of a European Design Organisation Approval, and how FAA designee-approved repair data are accepted on EU-registered engines is governed by the Technical Implementation Procedures; the research did not verify the relevant clause. [D4 U8]

41. The PMA market: HEICO, Chromalloy and others

The approved-alternative-parts business has one listed company large enough to be read from its filings, one private company that holds the CFM56 hot-section approvals, a handful of others, and a set of market-size estimates that do not agree. This section takes them in that order and ends with what the research found about how far alternative parts have penetrated each category of part.

41.1 HEICO FLIGHT SUPPORT GROUP

Part IV §29 glossed the company; this Part carries the full treatment, and Part XI §87 the peer comparison.

HEICO FLIGHT SUPPORT GROUP (FSG)

The Flight Support Group is the larger of HEICO Corporation's two operating groups and the largest independent PMA business. HEICO, listed on the NYSE, describes the group as one that "designs, manufactures, repairs, overhauls and distributes jet engine and aircraft component replacement parts, which are approved by the FAA", using "proprietary technology to design and manufacture jet engine and aircraft component replacement parts for sale at lower prices than those manufactured by OEMs", and describes itself as "the largest independent supplier of non-OEM jet engine and aircraft component replacement parts". For the fiscal year to 31 October 2025 the group reported net sales of \$3,117.3M and operating income of \$750.4M, an operating margin of 24.1% derived in the research. The company says it has been adding new approvals at "approximately 300 to 500" a year by one dossier's reading of the annual report, or "approximately 400 to 550" by another's. It acquired Wencor Group, the other large independent PMA and distribution business, in 2023. Its discount to the manufacturer's price is not disclosed in anything retrieved. [HEICO 10-K FY2025; HEICO Q4 FY2025 release, 18 Dec 2025; HEICO release on Wencor, 2023; D11 §3.4; reconciliation R-024]

EXHIBIT 6.5 — HEICO FLIGHT SUPPORT GROUP: THE FIGURES THE RESEARCH REACHED, WITH THE TWO READINGS OF THE APPROVAL RATE

Metric	FY2024 (to 31 Oct 2024)	FY2025 (to 31 Oct 2025)	Change	Source
Net sales	\$2,639.4M	\$3,117.3M	+18%	HEICO 10-K FY2025; Q4 FY2025 release, 18 Dec 2025
Operating income	\$593.1M	\$750.4M	+27%	Same
Operating margin (derived)	22.5%	24.1%	+1.6 points	Derived in the research from the two lines above
Aftermarket replacement parts, organic growth	—	+15%, + \$263.9M	—	Q4 FY2025 release
Q1 FY2025 operating margin (quarter to 31 Jan 2025)	—	23.3%	—	Q1 FY2025 release, Feb 2025
New PMAs added per year	"approximately 300 to 500" (D4's reading) / "approximately 400 to 550" (D11's reading)	Same filing, two readings	—	HEICO 10-K FY2025 (R-024)

Metric	FY2024 (to 31 Oct 2024)	FY2025 (to 31 Oct 2025)	Change	Source
Pricing statement	"lower prices than those manufactured by OEMs"; "the functional equivalent" of OEM parts	—	—	HEICO 10-K FY2025
Stated discount to OEM price; portfolio split by part category; cumulative PMA count	Not disclosed in the text retrieved	—	—	D4 U9; D11 U5

Source: HEICO Corporation, Form 10-K for the fiscal year ended 31 Oct 2025; HEICO Q4/FY2025 earnings release (8-K exhibit 99.1), 18 Dec 2025; HEICO Q1 FY2025 earnings release, Feb 2025; HEICO release on the acquisition of Wencor Group, 2023. Compiled in D4 §4.1 and D11 §3.4. Reconciliation R-024. Accessed 2026-10-03.

The approval-rate figures need a sentence. Two dossiers read the same annual report and returned two ranges, "approximately 300 to 500" and "approximately 400 to 550" new approvals a year. Both are readings of the FY2025 10-K; the research did not establish whether the filing contains both phrases in different places or whether one is a search-extraction error, and the primer carries both. [HEICO 10-K FY2025, via D4 and D11; reconciliation R-024] Either way the order of magnitude is hundreds of approvals a year across a portfolio dominated by less complex parts, which is the simple end of the interval §38.3 discussed.

WORKED EXAMPLE 6.3 — DERIVING THE MARGIN A MATURE ALTERNATIVE-PARTS BUSINESS REPORTS, AND ITS INCREMENT

Inputs: Flight Support Group net sales \$2,639.4M (FY2024) and \$3,117.3M (FY2025); operating income \$593.1M and \$750.4M (all sourced: HEICO 10-K FY2025 and the Q4 FY2025 release). The group also includes repair and distribution, so these are not pure PMA margins (sourced caveat: HEICO 10-K, via D4 §3).

$$\begin{aligned}
 \text{operating margin FY2024} &= 593.1 \div 2,639.4 = 22.5\% \\
 \text{operating margin FY2025} &= 750.4 \div 3,117.3 = 24.1\% \\
 \text{incremental margin FY2024 to FY2025} &= (750.4 - 593.1) \div (3,117.3 - 2,639.4) = 157.3 \div 477.9 = 32.9\%
 \end{aligned}$$

So the group earned 24.1% of sales as operating income in FY2025, after depreciation and amortisation, and 32.9% on the additional sales it added in the year; both figures are derived in the research. They are on an operating-income basis and are not comparable without adjustment with FTAI's segment Adjusted EBITDA margin (earnings before interest, taxes, depreciation and amortisation as FTAI adjusts them, Part X §80), which adds back depreciation; Part XI §88 sets the bases side by side. Limits of the example: the group mixes PMA with repair and distribution, whose margins the filing does not separate; HEICO's fiscal year ends in October; and the discount at which

its parts sell is undisclosed, so the margin cannot be decomposed into a price and a cost. [HEICO 10-K FY2025; HEICO Q4 FY2025 release; D11 §2.5, §3.4]

41.2 CHROMALLOY

Part IV §29 glossed the company. This Part carries it in full because it holds the approvals that matter most to the CFM56 and because of its relationship with FTAI (§43).

CHROMALLOY

Chromalloy is a Palm Beach Gardens, Florida, company that repairs and coats turbine-engine parts, makes investment castings (a casting method in which a ceramic mould is formed around a wax pattern of the part), and designs and manufactures PMA parts. Trade and deal coverage put its revenue at "\$1 billion-plus" and its headcount at about 4,000; Chris Celtruda was appointed chief executive in 2024. It is owned by Veritas Capital through its parent Sequa Corporation: Carlyle bought Sequa in 2007 and sold it to a Veritas affiliate in a transaction announced in September 2022 and closed in December 2022, after Bloomberg reported in July 2022 that Carlyle sought about \$2B. It was an early hot-section PMA maker through BELAC (its joint venture with Lufthansa Technik and United Airlines), which made a stage-1 high-pressure turbine blade for the CFM56-3 and its military version the F108, and it has held since 30 October 2025 what it calls "the only aftermarket alternative" to the manufacturer's new high-pressure turbine blade for the CFM56-5B/7B, with a stage-1 vane, a low-pressure turbine vane and, by one reading, a shroud approved alongside. Its CFM56 blade production was sold out for the rest of 2025 with slots being reserved into 2026. [GovConWire, Dec 2022; Bloomberg, 19 Jul 2022; FlightGlobal, Chromalloy profile, undated; PRNewswire, BELAC F108 contract, 2000s; Chromalloy release, 30 Oct 2025; D4 §3, §4.2; D5 §2.7]

Two distribution agreements show how its parts reach shops. AAR Corp signed an engine-parts supply agreement with Chromalloy in November 2024 and an exclusive agreement covering the PW4000 engine in March 2025. [PRNewswire, Nov 2024 and Mar 2025] Chromalloy has also announced an FAA-approved high-pressure turbine blade PMA for the V2500, the CFM56's competitor on the A320ceo, unveiled at an MRO Europe conference whose date the research did not capture. [Aviation Week, MRO Europe, date not captured; D4 U17]

The CFM56-5B/7B roster is the part of the record that matters most to this primer, and the sources disagree on its length.

EXHIBIT 6.6 — CHROMALLOY'S PMA ROSTER FOR THE CFM56 FAMILY, WITH DATES AS THE SOURCES GIVE THEM AND THE THREE-VERSUS-FOUR TENSION

Part	Engine	Approval date as given	What the source says	Source and level
High-pressure turbine blade	CFM56-5B/7B	Announced 30 Oct 2025	"the only aftermarket alternative to the OEM manufactured new part"; orders cover "all planned production for the rest of 2025"; "many customers now reserving manufacturing slots well into 2026"	Chromalloy release (company release); Newswire copy; AviTrader 31 Oct 2025 and Aviation Maintenance Magazine (trade press)
High-pressure turbine stage-1 vane (the HPT nozzle)	CFM56-5B/7B	Chromalloy release undated in the research; third-party commentary dates it to November 2025	FAA approval of "HPT Turbine Vane 1 PMAs"	Chromalloy release (company release); Komodo Capital (investor blog) for the date
Low-pressure turbine vane, described in the headline as an "up-change" PMA (what the term covers was not retrieved)	CFM56	Date not captured	FAA approval	Aeromorning (trade press, undated); D4 U17
High-pressure turbine shroud (the ring segments surrounding the blade tips, Part I §5)	CFM56-5B/7B	Not separately dated	Listed as one of "four PMA parts" Chromalloy "now offers" in one dossier's reading of the Newswire copy of the 30 Oct 2025 release	D5 §2.7 reading of the Newswire copy (company release as read by D5)
Stage-1 high-pressure turbine blade	CFM56-3 / F108	2000s	Made by BELAC LLC; US Air Force one-year contract of \$2.6M for F108 blades	PRNewswire (company release)

Part	Engine	Approval date as given	What the source says	Source and level
High-pressure turbine blade	V2500 (contrast)	Date not captured	FAA-approved, unveiled at MRO Europe	Aviation Week (trade press); D4 U17

Source: Chromalloy, "Chromalloy Secures FAA Approval of CFM56 High Pressure Turbine Blade PMA", 30 Oct 2025, and the Newswire.com copy; Chromalloy, "Chromalloy Receives FAA Approval of CFM56-5B/7B HPT Turbine Vane 1 PMAs", 2025; Aeromorning, LPT vane up-change PMA, undated; AviTrader, 31 Oct 2025; Aviation Maintenance Magazine, 2025; Aviation Week, MRO Europe; PRNewswire, BELAC F108 contract; D4 §4.2; D5 §2.7. Reconciliation R-025. Accessed 2026-10-03.

The count is contested. One dossier, reading Chromalloy's own announcements, found three CFM56-5B/7B approvals: the blade (30 October 2025), the stage-1 vane (November 2025 per commentary) and a low-pressure turbine vane (undated). Another dossier, reading the Newswire copy of the 30 October release, recorded that Chromalloy "now offers four PMA parts on the CFM56-5B/7B: the HPT blade, the HPT nozzle guide vane, the LPT stage-1 nozzle guide vane and the HPT shroud". [D4 §4.2; D5 §2.7; reconciliation R-025] Investor blogs add a third and fourth count: "two approved (HPT, LPT), three pending", and the HPT vane as "the second of five". [Kairos Research and Komodo Capital, investor blogs, undated; D4 T2] These are readings of one release family and of commentary on it, and the primer presents the three-versus-four count as unresolved. Whether the low-pressure turbine vane is one of the "five" the blogs describe is not stated by any source. [D4 T2]

41.3 OTHER MAKERS, AND THE MANUFACTURER THAT ONCE MADE PMA PARTS

Two other names appear in the research. Wencor Group was the other large independent PMA and distribution business until HEICO acquired it in 2023, so the independent industry's two largest portfolios are now under one owner. [HEICO release on Wencor, 2023] And in the late 2000s an engine manufacturer itself obtained alternative-part approvals for a competitor's engine: Pratt & Whitney ran a unit called Global Material Solutions that obtained FAA PMA for CFM56-3 parts, producing a public dispute with CFM at the Farnborough air show that FlightGlobal covered, along with P&W's parts tests. The dates and the unit's eventual wind-down were not re-verified, and the research found the episode only through headlines. [FlightGlobal, "Farnborough: P&W parts display fuels CFM spat" and "P&W to conclude PMA parts tests in six weeks", undated; D4 §3]

A vocabulary note belongs here because two dossiers used one phrase in two senses. "Material solutions", in this primer, means what Part IV §29 means by it: a shop's or manufacturer's package of used material, repaired parts and PMA parts assembled to reduce the new-parts bill. The research did not match the phrase to any current CFM or GE programme by that name; the only programme it found with those words was the historical Pratt & Whitney unit above. [D2 §8; D4 U10; registry C-19]

41.4 MARKET SIZE AND PENETRATION

Every market-size figure the research reached is from a market-research vendor or a trade article, and they do not agree with each other or, in one case, with themselves.

EXHIBIT 6.7 — PMA MARKET-SIZE AND PENETRATION ESTIMATES: THE CONFLICT EXHIBIT

Estimate	Figure	Scope as the source states it	Source and level	Date
Commercial-aircraft PMA market	\$11.8B in 2025, rising to \$16.1B in 2034 (3.37% compound annual growth rate, CAGR)	"Commercial aircraft PMA"; engines stated as the largest segment	IMARC (a market-research vendor)	2025 base
Commercial-aircraft PMA market	\$14B by 2026	"Commercial aircraft PMA"	Global Industry Analysts (GIA), via PRNewswire (market-research vendor)	2022 release
Engine PMA parts plus DER repair market	\$2,952.2M in 2022, rising to \$6,174.7M by 2032	Engine PMA and DER repair together	A vendor report listed by GII (a Korean listing site for market-research reports)	2022 base
PMA share of the engine-parts market	"capped at around 21%"	Engine parts	FlightGlobal (trade press), older article, undated	Undated
Global aircraft aftermarket parts	\$56.5B in 2026, rising to \$91.3B in 2033; engine parts 34.8%	All aftermarket parts, new and used, OEM and PMA	Persistence Market Research (market-research vendor)	2026 base
One airline's PMA saving	\$54M from 341 PMA part numbers in 2022	Southwest Airlines, fleet-wide	Aviation Week (trade press)	c. 2023

Source: IMARC Group, "Commercial Aircraft Parts Manufacturer Approval Market" (2025 base); PRNewswire / Global Industry Analysts, 2022; GII Korea listing, "Global Engine PMA Parts & DER Repair Market Research" (2022 base); FlightGlobal, "Aftermarket shocks", undated; Persistence Market Research, "Aircraft Aftermarket Parts Market" (2026–2033); Aviation Week, "Southwest's 737-700 Engine Strategy As It Phases Out Fleet", c. 2023. Compiled in D4 §4.4. Reconciliation R-023; D4 T4 and T7. Accessed 2026-10-03.

Three disagreements sit in that table and the primer carries all of them. The first is between the two commercial-aircraft figures: \$11.8B in 2025 from one vendor against \$14B by 2026 from another, from a release three years older. [IMARC, 2025; GIA via PRNewswire, 2022; reconciliation R-023] The second is between scopes: a vendor's engine PMA-plus-DER market of about \$2.95B in 2022 sits beside another vendor's statement that engines are the largest segment of an \$11.8B

total, which the two cannot both be unless the scopes differ in ways neither states. [GII listing, 2022 base; IMARC, 2025; D4 T4] The third is between a ceiling and a forecast: a trade article put PMA's share of engine parts at "capped at around 21%", while a vendor forecasts the engine PMA-and-DER market doubling between 2022 and 2032; both are forecasts and neither is sourced to disclosed data. [FlightGlobal, undated; GII listing; D4 T7] The scope definitions behind each figure were not retrieved, and the primer does not average or choose. [D4 §4.4]

Penetration by category of part is better described in words than in numbers, because no source gave a split. Consumables and expendables, the seals, bushings, fasteners, filters and brackets of §38.2, are where PMA has long been established and where HEICO's hundreds of approvals a year mostly fall; HEICO's own split of its portfolio by category was not disclosed. [D4 §2.3, U9] Rotating life-limited parts have zero penetration: no PMA exists for any CFM56 disk, spool or shaft. [D4 U7] The cooled hot-section airfoils sit between: the CFM56-3 blade of the 2000s through BELAC, and the CFM56-5B/7B blade, vane and possibly shroud of late 2025, are the approvals the research found, and the customer uptake behind them is known only as "sold out for 2025". [D4 §4.2, U15] The one operator-level datapoint, Southwest's 341 part numbers and \$54M in 2022, is fleet-wide and mostly non-hot-section, and §42.1 works it. [Aviation Week, c. 2023]

42. Acceptance: airlines, lessors, and the OEM's responses

A PMA part is legal to install the day it is approved. Whether it is installed depends on three other parties: the airline that owns or operates the engine, the lessor whose lease may forbid it, and the manufacturer whose warranty, service and residual-value programmes may discourage it. This section presents each party's position as the sources state it, with dates, and does not adjudicate between them.

42.1 AIRLINES

The operator-level evidence in the research is thin and is set out in full.

EXHIBIT 6.8 — AIRLINE DATAPOINTS ON PMA AND DER USE, AS THE RESEARCH FOUND THEM

Airline	Datapoint	Date	Source and level
Southwest Airlines	341 PMA part numbers in use; \$54M saved in the year; evaluating PMA and DER options for the CFM56-7B fleet as the 737-700s are phased out	2022 figures; article c. 2023	Aviation Week (trade press)

Airline	Datapoint	Date	Source and level
United Airlines	Launch customer for BELAC's CFM56-3 stage-1 HPT blade PMA; a partner in the BELAC venture	2000s	PRNewswire (company release) and search summary
Delta, American, Ryanair and non-US CFM56 operators	No PMA-policy source retrieved for hot-section parts	—	D4 U11
Unnamed customers of Chromalloy	Orders covering "all planned production for the rest of 2025"; "many customers now reserving manufacturing slots well into 2026"	30 Oct 2025	Chromalloy release (company release); customers unnamed (D4 U15)

Source: Aviation Week, "Southwest's 737-700 Engine Strategy As It Phases Out Fleet", c. 2023; PRNewswire, BELAC F108 contract release, 2000s; Chromalloy release, 30 Oct 2025; D4 §3 and U11, U15. Accessed 2026-10-03.

WORKED EXAMPLE 6.4 — SOUTHWEST'S SAVING PER PART NUMBER, AND WHY IT CANNOT BE CONVERTED TO A PER-VISIT FIGURE

Inputs: \$54M saved in 2022 from 341 PMA part numbers (sourced: Aviation Week, c. 2023). A fleet of roughly 800 737s (as the dossier states it; unsourced in the research).

$$\begin{aligned}\text{saving per part number per year} &= 54,000,000 \div 341 = 158,358 \\ \text{saving per aircraft per year, at 800 aircraft} &= 54,000,000 \div 800 = 67,500\end{aligned}$$

So the airline saved about \$158,000 per approved part number per year, or about \$67,500 per aircraft per year on the fleet figure the dossier used. Neither figure converts to a saving per shop visit, because the 341 part numbers span airframe and engine, consumables and components, and the article does not give the part mix or the number of shop visits in the year. The figure is a fleet-wide, mostly non-hot-section datapoint from before any CFM56-5B/7B hot-section PMA existed, and §44 does not use it except as a cross-check of scale. Limits of the example: the fleet size is unsourced; the saving's basis (against list price, or against what the airline would otherwise have paid) is not stated.

[Aviation Week, c. 2023; D4 §2.8]

One episode in 2023 should be named here so that it is not confused with the subject of this Part. In 2023 several US airlines, including Delta, United, Southwest and American, found parts with forged documentation in CFM56 engines, supplied through a London broker called AOG Technics (the abbreviation AOG means aircraft on ground, the trade's term for an urgent parts need), which then faced a lawsuit. Those were unapproved parts passed off with false release paperwork. PMA parts are FAA-approved parts with genuine release paperwork, and the two are different categories. [Fox Business, 2023; D4 §3]

42.2 LESSORS

An operating lease specifies the state in which the aircraft or engine must be returned, and those return conditions, which Part II §13 treats in full and Part I §9 glossed, are where a lessor's view of alternative parts is written down. One lease-management primer calls return conditions "the most important part of any lease", since they, not the rent, protect the lessor's next placement.

[Sofema Online, undated]

OEM-ONLY CLAUSE (NO-PMA / NO-DER CLAUSE; TCH-CONFIGURATION CLAUSE)

An OEM-only clause is a lease return condition requiring that at redelivery the engine contain only parts produced by the type-certificate holder and only repairs from the type-certificate holder's manuals. A lessee that used PMA parts or DER repairs during the term must then either remove them at the last heavy check before return or pay compensation. The sources describe both the clause and the workaround: "when permitted by the lease, lessees can ... use PMA throughout the term of the lease and ... remove PMA parts during the heavy check prior to the return", while for engines "returning a 'PMA-heavy' engine to strict TCH configuration carries significant financial burdens". A lease that requires TRUEngine status (Part I §3) is an OEM-only clause by another name. Engine lessors were reported to forbid PMA and DER "in the vast majority of cases" in 2016; aircraft-lease clauses were reported to be loosening in 2015; practice in 2024–2026 is unknown. [Aviation Suppliers Association blog, undated; ELFC, mid-2016; MARPA, 10 Nov 2015; D4 §2.9, U1]

EXHIBIT 6.9 — LESSOR POSITIONS ON PMA AND DER, BY DATE, ASSET CLASS AND SOURCE

Date	Statement as the source gives it	Asset class	Source and level
2009	Operating lessors "most often stipulated the use of OEM parts in the return conditions"	Aircraft leases	Aircraft Commerce, Issue 67 (trade press)
10 Nov 2015	"most lessors permit PMA to be used on leased aircraft (with a few exceptions); the carrier just has to demand the right"; lessors "grow more willing to waive the 'no PMA' clauses"	Aircraft leases	MARPA, the Modification and Replacement Parts Association (a PMA trade association)
Mid-2016	For engines, "lessors, in the vast majority of cases support the OEMs by forbidding the installation of PMA and DER"; "returning a 'PMA-heavy' engine to strict TCH configuration carries significant financial burdens"	Engine leases	Engine Lease Finance Corporation (ELFC), an engine lessor (company article)
Undated	"when permitted by the lease, lessees can ... use PMA throughout the term of the lease and ... remove PMA parts during the heavy check prior to the return"	Aircraft leases	Aviation Suppliers Association blog (a parts-trade association)

Date	Statement as the source gives it	Asset class	Source and level
Undated	PMA parts "continue to undermine full-life maintenance adjustments"	Engine appraisal	Aircraft Value News (trade press), via Parts I and V
2024–2026	No statement retrieved from AerCap, Air Lease, Avolon, SMBC Aviation Capital, BOC Aviation, Willis Lease or any engine lessor	—	D4 U1
FY2025	FTAI lists lessors among the airline, lessor and maintenance customers to whom its PMA portfolio offers "cost savings and flexibility"	—	FTAI 10-K FY2025 (filing)

Source: Aircraft Commerce, Issue 67, 2009 (maintenance article on PMA and lessors); MARPA, "Are leases getting looser?", 10 Nov 2015; Engine Lease Finance Corporation, "An overview of the aircraft engine aftermarket in mid 2016"; Aviation Suppliers Association, "Airlines: now is the time to choose lessors who accommodate PMA and DER repairs", undated; Aircraft Value News, "PMA continues to undermine full-life maintenance adjustments", undated; FTAI 10-K FY2025; D4 §2.9, §3 and U1. Reconciliation D4 T6. Accessed 2026-10-03.

Two of those statements pull in opposite directions, and the difference is partly one of asset class and partly one of date. The 2015 statement that "most lessors permit PMA ... with a few exceptions" concerns aircraft leases and comes from a PMA trade association. The 2016 statement that engine lessors forbid PMA and DER "in the vast majority of cases" concerns engine leases and comes from an engine lessor. [MARPA, 10 Nov 2015; ELFC, mid-2016; D4 T6] Both can be true at once, and the primer presents them as such without inferring either onto the other asset class.

What the research could not find is any statement of current practice. Every lessor position retrieved dates from 2009 to 2016. No lessor's 2024–2026 annual report, lease description, survey response or conference remark on PMA or DER clauses was reached, so whether the loosening reported in 2015 continued, reversed or reached engine leases is unknown, and the primer says so rather than extrapolating. [D4 U1] Part II §13 describes return conditions in detail without any statement on PMA clauses, which is consistent with the gap rather than evidence either way. The mechanism by which the clause bears on value is documented without being quantified: an appraiser's full-life maintenance adjustment assumes manufacturer-configuration parts, and Aircraft Value News records that PMA content "continue[s] to undermine" that adjustment; no dollar or percentage effect on an engine or aircraft appraisal was found. [Aircraft Value News, undated; D4 U13; Part I §8]

42.3 THE MANUFACTURER'S RESPONSES

CFM International's recorded responses fall into three kinds: commitments it made in 2018 about warranty and service, a configuration designation it markets to lessors and appraisers, and commercial responses in supply, repair and agreements. Each is presented as CFM states it.

CONDUCT POLICIES (CFM–IATA, 2018)

The Conduct Policies are the commitments CFM International published in 2018 to settle a complaint the International Air Transport Association (IATA) had brought to the European Commission about CFM's aftermarket conduct. Under them, installing non-CFM parts or repairs "does not in itself render the warranty void"; CFM honours its warranty on its own parts and repairs in an engine that contains non-CFM parts; and CFM "does not refuse to service engines because they contain non-OEM parts or repairs". The policies also cover CFM's licensing of its shop manual to third-party shops and its sale of CFM parts to them; the research did not re-read those clauses. Trade associations summarised the agreement as CFM agreeing "to permit the use of PMA parts and DER repairs". [IATA, "IATA–CFM Agreement on Engine Maintenance", 2018; IATA, "CFM Conduct Policies and Implementation Measures"; CFM, "External Guidance", Mar 2019; MARPA, 1 Aug 2018; Aviation Suppliers Association bulletin, Oct 2018]

The second response is the designation Part I §3 defined. TRUEngine designates an engine as maintained to CFM-approved configuration, meaning CFM parts and CFM-approved repairs; CFM markets it as supporting residual value and resale, extended it in 2012 to cover subsequent buyers, and lessors can require it in return conditions. [GE Aerospace, TrueEngine presentation; CFM release on the TRUEngine expansion; Aviation International News, 12 Sept 2012] The 2018 policies and the designation are both CFM positions and address different levers: one is a contractual commitment about warranty and service, the other a marketing statement about resale value. The primer records both without treating either as the "real" position. [D4 T8]

The third kind is commercial, and the dated items are in the exhibit.

EXHIBIT 6.10 — CFM INTERNATIONAL'S COMMERCIAL RESPONSES IN THE CFM56 HOT SECTION AND TO INDEPENDENTS, DATED

Date	Response	Figures as CFM gives them	Source and level
2018	Conduct Policies settling IATA's complaint to the European Commission	No warranty voiding for non-CFM parts or repairs; no refusal of service; shop-manual licensing and parts sales to third-party shops covered	IATA and CFM documents (association and company)

Date	Response	Figures as CFM gives them	Source and level
2012 (expansion)	TRUEngine designation extended to subsequent buyers	Configuration-assurance designation marketed for residual value	CFM / GE release; Aviation International News
16 Oct 2023	Upgraded CFM56-5B/7B HPT blade introduced	Increased wall thickness, optimised dovetail loading, tightened tolerances; "up to 25 percent longer life in certain operating environments"	Safran / CFM release, 16 Oct 2023 (company release)
2025/26, date not captured	HPT blade output and repair capacity raised	1,000th and later 1,700th set of new blades shipped; output "almost two and a half times" year over year; about 400 CFM56 repairs industrialised since early 2023, including automated laser welding of HPT blades that cuts repair turnaround by about a third	CFM stories, "Investing in Durability" and "CFM boosts HPT blade output and repair" (company articles)
Same	Positioning statement	The upgraded blades are "designed to keep customers flying with OEM parts they know and trust" and "can help extend time on-wing to optimize cost of ownership and enhance residual value" (Jacey Welsh, CFM executive vice president for CFM56)	CFM story (company article)
FY2025	Material supply to independents	GE's priority-supplier material input up more than 40% year over year under its operating system; "an open network exists with multiple providers who compete and invest to win shop visits"; GE will "continue to strengthen MRO access to OEM materials to support further CFM56 longevity"	GE 10-K FY2025; GE Aerospace, 2026 (filing and company statement)
22 Jan 2026	Multi-year materials agreement with FTAI Aviation	FTAI "secures OEM replacement part supply, thrust performance upgrades and component repair"; both parties describe it as strengthening "the open maintenance, repair, and overhaul (MRO) ecosystem"; terms undisclosed	FTAI release, GlobeNewswire, 22 Jan 2026 (company release)

Date	Response	Figures as CFM gives them	Source and level
—	Statement on Chromalloy's CFM56 HPT blade PMA	None found	D4 U10

Source: IATA, 2018 agreement and Conduct Policies; CFM External Guidance, Mar 2019; GE Aerospace TrueEngine presentation; CFM TRUEngine expansion release and Aviation International News, 12 Sept 2012; Safran / CFM, "CFM introduces upgraded HPT blade for CFM56 engines", 16 Oct 2023; CFM International, "Investing in Durability: CFM ships 1,000th set of new CFM56 HPT blades" and "CFM boosts HPT blade output and repair to meet extended demand for CFM56 engines", 2025/26; GE Form 10-K FY2025; GE Aerospace, 2026, via D2; FTAI Aviation, "FTAI Aviation Announces Multi-Year Materials Agreement with CFM International", 22 Jan 2026. Compiled in D4 §3 and D2 §2.3. Accessed 2026-10-03.

The redesigned blade of October 2023 and the alternative blade of October 2025 are two answers to the same shortage, and the manufacturer's figures for its own answer are these. CFM introduced an upgraded CFM56-5B/7B high-pressure turbine blade on 16 October 2023 with increased wall thickness, optimised dovetail loading and tightened tolerances, claiming "up to 25 percent longer life in certain operating environments". [Safran / CFM release, 16 Oct 2023] It later reported shipping the 1,000th and then the 1,700th set of new blades, with output "almost two and a half times" the prior year's, and about 400 CFM56 repairs industrialised since early 2023. [CFM International stories, 2025/26, dates not captured] A longer-lived blade and a faster blade repair both extend the window in which a blade is repaired rather than replaced, which under the cadence in §40.3 is the window in which no new blade, manufacturer's or alternative, is bought. [D4 §5] The research found no CFM statement about the Chromalloy approval itself. [D4 U10]

The last row needs its own definition, because the agreement is one of the levers FTAI's management named (§43) and Part VII §52 owns its treatment within the segment.

CFM MATERIALS AGREEMENT (FTAI-CFM, 22 JANUARY 2026)

The CFM materials agreement is the multi-year agreement FTAI Aviation announced on 22 January 2026 under which, in the release's words, FTAI "secures OEM replacement part supply, thrust performance upgrades and component repair" from CFM International, described by both parties as strengthening "the open maintenance, repair, and overhaul (MRO) ecosystem". Its duration, volumes, pricing, and whether it bears on FTAI's use of PMA parts, were not disclosed in anything retrieved. Part VII §52 carries the full treatment within the Aerospace Products segment. [FTAI release, GlobeNewswire, 22 Jan 2026; D4 U14; reconciliation K-19]

On the Q4 2025 call FTAI's chief executive described it as "a materials agreement with CFM that includes parts and repairs". [MarketBeat call coverage, 26 Feb 2026] That the same company holds, at once, an agreement to buy the manufacturer's new parts and repairs and a joint venture to make

alternative parts is a fact the sources record; what the agreement says about the alternative parts is not. [D4 U14]

43. The FTAI–Chromalloy joint venture: what is filed and what is reported

FTAI's parts venture is described at three levels of source that say different things, and this section keeps them apart. Part IV §29 glossed the venture; this Part carries what is filed and what is reported, and Part VII §52 carries its effect within the segment.

FTAI–CHROMALLOY JOINT VENTURE (THE PMA JOINT VENTURE)

FTAI's FY2025 annual report says only that the company "develops and manufactures Parts Manufacturer Approval ('PMA') parts through a joint venture", as part of a "proprietary portfolio of products that enables it to provide cost savings and flexibility to airline, lessor, and maintenance, repair, and operations customers"; the partner, the ownership split, the accounting treatment, the revenue and the savings are not in the filing text retrieved. The orchestrator's reading of the same annual report's investments note lists an "Advanced Engine Repair JV" held at 25%, with \$15.0M invested in December 2016 and \$13.5M in August 2019 and a carrying value of \$22,429K at 31 December 2025, and records that whether this is the PMA joint venture is unknown. Third-party commentary describes an exclusive, perpetual partnership with Chromalloy formed in 2018 covering five CFM56 hot-section parts supplied to FTAI "at cost", with savings "up to \$2 million per shop visit"; none of that is in FTAI's filings as retrieved. [FTAI 10-K FY2025; FTAI 10-Q Q2 2026; orchestrator notes on the FY2025 10-K investments note and the Q3 2025 10-Q note; In Practise, Komodo Capital, Kairos Research, Yahoo Finance bull-case digest (third-party commentary); reconciliation R-026]

43.1 THREE LEVELS OF SOURCE

EXHIBIT 6.11 — THE FTAI–CHROMALLOY RELATIONSHIP AT THREE SOURCE LEVELS

Item	FTAI's filings (10-K FY2025; 10-Q Q2 2026)	Orchestrator's reading of the FY2025 10-K investments note	Third-party commentary (In Practise; Komodo Capital; Kairos Research; Yahoo Finance digest)
Existence	"develops and manufactures Parts Manufacturer Approval ('PMA') parts through a joint venture"	An "Advanced Engine Repair JV" among the equity-method investments; identity with the PMA venture unknown	An "exclusive perpetual partnership" with Chromalloy

Item	FTAI's filings (10-K FY2025; 10-Q Q2 2026)	Orchestrator's reading of the FY2025 10-K investments note	Third-party commentary (In Practise; Komodo Capital; Kairos Research; Yahoo Finance digest)
Partner	Not named in the text retrieved	Not named	Chromalloy
Formation	Not retrieved	\$15.0M invested Dec 2016; \$13.5M more Aug 2019	2018
Ownership and structure	Not retrieved	25%; equity method	A supply partnership; parts supplied "at cost"
Carrying value	Not retrieved	\$22,429K at 31 Dec 2025	—
Parts covered	Not retrieved	—	Five CFM56 hot-section parts
Approved to date	Not retrieved	—	"two approved (HPT, LPT), three pending" (one blog); the HPT vane "the second of five" (another)
Savings	"cost savings and flexibility" (qualitative)	—	"up to \$2 million per shop visit"; "5–10 percentage points of additional margin"
Date mismatch	—	2016 first investment	2018 formation: inconsistent with the note's date if the two describe one entity

Source: FTAI Aviation Ltd., Form 10-K for FY2025 and Form 10-Q for the quarter ended 30 Jun 2026 (sec.gov); orchestrator notes, "Equity-method investments (10-K FY2025; 10-Q Q3 2025 note R13)"; In Practise, "FTAI, Chromalloy and CFM56 HPT Blade PMA" and "FTAI & PMA Parts"; Komodo Capital (Substack); Kairos Research (Substack); Yahoo Finance, "FTAI Aviation Ltd. (FTAI): A Bull Case Theory". Compiled in D4 §4.3 and the orchestrator notes. Reconciliation R-026 and R-027; registry 1c. Accessed 2026-10-03.

The three columns describe different things. A filing describes "a joint venture" through which FTAI "develops and manufactures" parts, which is the language of an equity interest in a manufacturing entity. Commentary describes a "partnership" under which parts are supplied "at cost", which is the language of a supply agreement. The orchestrator's reading of the investments

note describes a 25% equity-method stake first funded in December 2016, two years before the commentary's formation date. If the three describe one arrangement, the structure (equity venture against supply agreement) and the date (2016 against 2018) are both unreconciled; if they describe two, the filing's "joint venture" and the note's "Advanced Engine Repair JV" may be different entities. The research did not resolve either, and the primer does not. [reconciliation R-026; registry 1c] The chain map and one dossier call the venture "a joint venture with Chromalloy"; that is an orchestrator-level description that no filing retrieved supports by name. [D10 §3; orchestrator notes]

The investments note the orchestrator read lists other equity-method stakes, and seeing the list is the only way to see what the "Advanced Engine Repair JV" sits beside.

EXHIBIT 6.12 — FTAI'S EQUITY-METHOD INVESTMENTS AS THE ORCHESTRATOR READ THEM FROM THE FY2025 10-K AND THE Q3 2025 10-Q

Investment	Stake	Invested	Carrying value, 31 Dec 2025	What it is	Where treated
Advanced Engine Repair JV	25%	\$15.0M (Dec 2016) + \$13.5M (Aug 2019)	\$22,429K	Unknown whether this is the Chromalloy PMA joint venture or a separate repair-development venture	This Part §43; Part VII §52; Part X §79
2025 Partnership (Strategic Capital)	19% (FY2025 10-K) / 20% (Q3 2025 10-Q)	\$151.6M in 9M 2025	\$281,740K	The first Strategic Capital vehicle; FTAI is Servicer	Part VIII §61–64
QuickTurn Europe (Rome Fiumicino)	50%	\$10.5M (closed 5 Jun 2025)	\$9,987K	Engine shop joint venture, ex-IAG Engine Center	Part VII §50
Prime Engine Accessories (Bristol, Connecticut, with Bauer)	50%	Not returned	Not returned	Accessory-repair venture; "up to \$75,000 in average savings per shop visit" (Q3 2025 call)	Part VII §50

Source: orchestrator notes, "Equity-method investments (10-K FY2025; 10-Q Q3 2025 note R13)", citing FTAI Form 10-K FY2025 and the Q3 2025 10-Q; Insider Monkey, FTAI Q3 2025 call transcript, Oct 2025 (for the \$75,000 figure, via D5); GlobeNewswire, 5 Jun 2025 (QuickTurn Europe). Reconciliation K-04, K-07, R-048. Accessed 2026-10-03.

The register's instruction is followed here: the Advanced Engine Repair JV is reported as an equity-method investment, its identity with the PMA venture is stated as unresolved, and the two are not equated. [registry 1c; reconciliation K-04] The investments note of the FY2018 to FY2025 annual reports, and any 8-K on the venture, would resolve the question; the research did not read them. [D4 U3]

43.2 THE APPROVALS TIMELINE

EXHIBIT 6.13 — DATED EVENTS BEARING ON FTAI'S PMA PARTS, WITH THE SOURCE LEVEL OF EACH DATE

Date	Event	Source and level
Dec 2016	\$15.0M invested in the "Advanced Engine Repair JV" (if the same entity)	FY2025 10-K investments note, via orchestrator notes (filing)
2018	FTAI–Chromalloy "exclusive perpetual partnership" formed, covering five CFM56 hot-section parts	Komodo Capital; Kairos Research (investor blogs)
1 Aug 2018	CFM's Conduct Policies published, settling IATA's complaint	MARPA; IATA (association)
Aug 2019	\$13.5M more invested in the "Advanced Engine Repair JV" (if the same entity)	FY2025 10-K, via orchestrator notes (filing)
16 Oct 2023	CFM introduces its upgraded CFM56-5B/7B HPT blade	Safran / CFM release (company release)
30 Oct 2025	Chromalloy announces FAA PMA for the CFM56-5B/7B HPT blade; production sold out for 2025, slots into 2026	Chromalloy release; AviTrader; Aviation Maintenance Magazine (company release; trade press)
Nov 2025	HPT stage-1 vane PMA (date per commentary; Chromalloy's release undated in the research)	Chromalloy release; Komodo Capital (company release; investor blog)
Undated	LPT vane "up-change" PMA; HPT shroud (per one reading of the 30 Oct release)	Aeromorning (trade press); D5 reading of the Newswire copy
22 Jan 2026	FTAI announces the multi-year materials agreement with CFM	FTAI release, GlobeNewswire (company release)
26 Feb 2026	On the Q4 2025 call, the chief executive names the PMA HPT blade, more used material and the CFM materials agreement as the levers to "approach 40%" segment margin	MarketBeat call highlights; Motley Fool transcript (call coverage)

Source: as listed in each row; orchestrator notes (equity-method investments); Chromalloy releases, Oct–Nov 2025; Safran / CFM, 16 Oct 2023; FTAI release, 22 Jan 2026; MarketBeat, "FTAI Aviation Q4 earnings call highlights", 26 Feb 2026; Motley

Fool, FTAI Q4 2025 call transcript, 26 Feb 2026. Compiled from D4 §4.2–4.3, D5 §2.7 and the orchestrator notes. Accessed 2026-10-03.

43.3 "SOLD OUT THROUGH 2025" AND THE SLOT STATEMENTS

Chromalloy's announcement of 30 October 2025 said that orders covered "all planned production for the rest of 2025" and that "many customers [were] now reserving manufacturing slots well into 2026", without naming any customer. [Chromalloy release, 30 Oct 2025; Newswire copy; AviTrader, 31 Oct 2025] In this primer's vocabulary that is a production slot, a reserved place in a parts maker's output, which is distinct from the induction slot of an engine shop (Part IV §27) and the manufacturing-slot reservation of a turbine maker (Part IX). [registry C-18] Two things the statement does not give are the ones that would size it: Chromalloy's production capacity in blade sets a year, and FTAI's share of that capacity. [D4 U12] The limiting process is single-crystal casting and coating, which §38.2 named; no capacity figure was disclosed. [D4 §5] Customer uptake beyond "sold out" was not found: no named airline or shop, and no count of sets delivered. [D4 U15] If the arrangement is exclusive for the five parts, as commentary alone says, then other shops' access to those parts, and at what price, is a constraint on market-wide penetration that the sources do not describe. [D4 §5, U3]

43.4 WHAT FTAI'S MANAGEMENT HAS SAID

The company's own quantified statements about the venture are few, and they are set beside the commentary's figures so that the gap is visible.

On the Q4 2025 earnings call of 26 February 2026, asked how the Aerospace Products segment's margin would move from the mid-30s toward 40%, chief executive Joe Adams "cited approval of a PMA high-pressure turbine blade as a key factor, along with expanded access to lower-cost parts supply, including used serviceable material and a materials agreement with CFM that includes parts and repairs"; the company said it "can approach 40% aerospace margins in 2026 via parts approvals, lower-cost supply and expanded repair capabilities". [MarketBeat call highlights, 26 Feb 2026; D5 §2.7] The segment's Q4 2025 Adjusted EBITDA was \$195M at a 35% margin, and FY2025 segment Adjusted EBITDA was \$671.3M on segment revenue of \$1,936.2M, a derived 34.7%. [FTAI Q4/FY2025 release, 25 Feb 2026; FTAI 10-K FY2025 via orchestrator notes; D5 §4.1] No dollar figure for the PMA saving per shop visit or per module, no PMA share of parts consumed, and no equity-method earnings line for the venture were retrieved. [D5 U11] The one FTAI-quantified savings lever found anywhere in the research is the Bristol accessories venture, "expected to deliver up to \$75,000 in average savings per shop visit", said on the Q3 2025 call. [Insider Monkey, FTAI Q3 2025 call transcript, Oct 2025; D5 §2.6]

Against that, investor blogs state "up to \$2 million per shop visit" from five parts supplied at cost, and "5–10 percentage points of additional margin". [Komodo Capital; Kairos Research; Yahoo

Finance digest; reconciliation R-027] The reconciliation records this as a confidence mismatch: a blog dollar figure against a qualitative management statement. [reconciliation R-027] §44 tests the dollar figure against the parts arithmetic; the worked example here only places the two claims against the segment's reported scale.

WORKED EXAMPLE 6.5 — PLACING THE CLAIMED FIGURES AGAINST THE SEGMENT'S REPORTED SCALE

Inputs: FY2025 Aerospace Products segment revenue \$1,936.2M, of which products revenue \$1,600.5M and MRE (Maintenance, Repair and Exchange) Contract revenue \$335.8M; segment Adjusted EBITDA \$671.3M (all sourced: FTAI 10-K FY2025 via orchestrator notes; Part VII §53). Products revenue per module produced \$2.11M (FY2025) and \$2.34M (Q2 2026), derived in the research on the products-revenue numerator, carried as a bound and not a price (reconciliation R-046). Claimed figures: "5–10 percentage points of additional margin" and "up to \$2 million per shop visit" (sourced: investor blogs). FTAI's one quantified lever: "up to \$75,000 in average savings per shop visit" from the Bristol venture (sourced: Q3 2025 call, via Insider Monkey).

5 to 10 margin points on FY2025 segment revenue: $0.05 \times 1,936.2 = 96.8$
to $0.10 \times 1,936.2 = 193.6$ (\$M)
5 to 10 margin points on FY2025 products revenue: $0.05 \times 1,600.5 = 80.0$
to $0.10 \times 1,600.5 = 160.1$ (\$M)
claimed per-visit saving against products revenue per module: $2.0 \div 2.11$
= 95%; $2.0 \div 2.34 = 85\%$
claimed per-visit saving against FTAI's quantified lever: $2,000,000 \div$
75,000 = 27 times

So the blogs' margin claim, applied to FY2025 revenue, would be about \$80–194M a year depending on the base; and the blogs' per-visit saving is of the same order as the revenue FTAI books per module produced, and 27 times the only per-visit saving FTAI itself has quantified. None of this tests whether the claims are right; it shows their scale against filed figures. Limits of the example: the margin claim's base and period are not stated by the blogs; revenue per module is a bound with the five limits Part VII §56 lists; the blade approval came on 30 October 2025, so no FY2025 figure can contain its effect; and a per-visit saving applies only to visits where blades are replaced rather than repaired (§40.3). [FTAI 10-K FY2025 via orchestrator notes; D5 §2.11, §4.1; Insider Monkey, Oct 2025; Komodo Capital; Kairos Research; reconciliation R-027, R-046]

44. Worked example: the parts bill with and without PMA

This section does what the Part has been building toward: it puts a CFM56-7B performance restoration's hot-section parts bill on the page, prices it with the manufacturer's new parts and then with alternative parts, shows the saving as a function of a discount no source discloses, and

tests the claimed "\$2 million per shop visit" against the arithmetic. Every tension that touches the numbers is presented in full, and nothing is resolved.

44.1 THE SETTING AND THE INPUTS

The visit is a performance restoration, in which the core is opened and the high-pressure compressor, combustor and high-pressure turbine are restored to recover exhaust-gas temperature (EGT) margin (Part I §6). The cadence from §40.3 decides whether new blades are bought at all: at a first visit, around 20,000 cycles, blades are typically repaired; at a second, around 30,000, they typically cannot be and new ones are needed. [In Practise, undated, interview digest] The example is therefore a second or later visit, which under Part III's context is exactly the visit an engine kept past its planned retirement undergoes. Alternative parts touch only the on-condition hot-section parts of §38.1; they touch no life-limited part, so the life-limited-part line, 40–60% of parts cost when replaced, is unchanged in every case below. [Safe Fly Aviation, 2026, consultancy blog; D4 U7]

EXHIBIT 6.14 — INPUTS TO THE PARTS-BILL EXAMPLE, WITH THE PROVENANCE AND STATUS OF EACH

Input	Value	Source and level	Status
New manufacturer's CFM56 HPT blade, price per blade	About \$20,000	In Practise expert interview (interview digest), undated	Sourced; third-party commentary
Alternative per-blade listing	About €35,000 (about \$38,000–39,000 at the conversion the dossier applied; rate not stated)	Web page on a university host (ATU), undated	Weak provenance; condition (new or serviceable) unclear
HPT blades per CFM56-7B rotor	80	Commonly cited; not confirmed by any source retrieved (D4 U16)	Parameter, unverified
New HPT blade and vane set	"up to a few million dollars"	In Practise, undated	Sourced; taken as \$2.5–3.0M (assumed within the phrase)
HPT blades and vanes as a share of a visit's parts cost	15–25%	Safe Fly Aviation (consultancy blog), 2026	Sourced; contradicts the per-blade route (R-015)
Material as a share of a visit	60–70%	Air Cargo Week (trade press), 2025/26	Sourced

Input	Value	Source and level	Status
Performance-restoration cost, two ranges	\$1.2–1.6M; \$1.8–2.5M	Safe Fly Aviation (consultancy blog), 2026, two pages	Sourced; both carried (R-013)
PMA discount to the manufacturer's price	Not disclosed	No source; neither Chromalloy nor FTAI published a price (D4 U2)	Shown as a function at 20%, 30%, 40% (assumed)
Claimed FTAI saving from five hot-section parts	"Up to \$2 million per shop visit"	Komodo Capital; Kairos Research (investor blogs)	Not in FTAI filings retrieved (R-027)
Life-limited parts as a share of parts cost	40–60%	Safe Fly Aviation, 2026	Sourced; untouched by PMA (D4 U7)

Source: In Practise, "FTAI, Chromalloy and CFM56 HPT Blade PMA", undated; web listing of weak provenance (D4 source 72); Safe Fly Aviation, 2026, consultancy blog; Air Cargo Week, "The true cost of engine maintenance", 2025/26; Komodo Capital and Kairos Research, investor blogs, undated. Compiled in D4 §2.8 and §4.5. Reconciliation R-013, R-015, R-027. Accessed 2026-10-03.

44.2 THE BILL WITH THE MANUFACTURER'S NEW PARTS, AND THE BLADE-PRICE TENSION

WORKED EXAMPLE 6.6 — A CFM56-7B PERFORMANCE RESTORATION'S HPT PARTS BILL: MANUFACTURER'S NEW PARTS, THEN PMA PARTS AT AN UNDISCLOSED DISCOUNT

Step 1, the manufacturer's blade set, priced per blade. Inputs: 80 blades (unverified parameter) at \$20,000 (sourced: interview digest) or at about \$38,500 (sourced: weak web listing, condition unclear, dossier's conversion of €35,000).

at the interview price:	$80 \times 20,000 = 1,600,000$
at the listing price:	$80 \times 38,500 = 3,080,000$

Step 2, the same blades and vanes priced from the consultancy shares instead. Inputs: HPT blades and vanes 15–25% of parts cost (sourced: Safe Fly Aviation); material 60–70% of a visit (sourced: Air Cargo Week); visit \$1.2–1.6M or \$1.8–2.5M (sourced: Safe Fly Aviation, two pages). Part I §7.2 applied the share to the visit total as an approximation; the stricter reading applies it to the material line.

share of the visit total (Part I's approximation): $0.15 \times 1,200,000 = 180,000$ to $0.25 \times 1,600,000 = 400,000$; $0.15 \times 1,800,000 = 270,000$ to $0.25 \times 2,500,000 = 625,000$

share of the material line: $0.15 \times 0.60 \times 1,200,000 = 108,000$ to $0.25 \times 0.70 \times 1,600,000 = 280,000$; $0.15 \times 0.60 \times 1,800,000 = 162,000$ to $0.25 \times 0.70 \times 2,500,000 = 437,500$

This is the tension in full. Priced per blade, the blade set alone is \$1.6M to \$3.1M, which is 100% to 260% of a \$1.2–1.6M restoration and 64% to 171% of a \$1.8–2.5M one. Priced as a share, the blades and vanes together are \$0.11M to \$0.63M. The two routes cannot both describe the same visit at the same prices. The research recorded the possible reasons without establishing any: the 80-blade count is a recollection; the per-blade prices come from an interview and a listing of unclear condition rather than a catalogue; and a share figure drawn from visits at which blades are repaired rather than replaced would be far smaller than a visit at which the whole set is bought new. The primer carries both routes and stops (R-015). [D4 §2.8, T5, U16; Safe Fly Aviation, 2026; Air Cargo Week, 2025/26; Part I §7.2; reconciliation R-015]

Step 3, the PMA blade set at a discount d to the manufacturer's price, on the \$1.6M set. The discount is undisclosed (D4 U2); the three rates are illustrative assumptions.

Assumed PMA discount d	PMA set cost (\$1.6M base)	Saving on the blade set	PMA set cost (\$3.08M base)	Saving on the blade set
20%	\$1,280,000	\$320,000	\$2,464,000	\$616,000
30%	\$1,120,000	\$480,000	\$2,156,000	\$924,000
40%	\$960,000	\$640,000	\$1,848,000	\$1,232,000

Step 4, adding the stage-1 vane set. If blades and vanes together are "up to a few million dollars" (sourced phrase), taken as \$2.5–3.0M (assumed within the phrase), the same discounts give savings on the two part families together of:

at 20%: $0.20 \times 2,500,000 = 500,000$ to $0.20 \times 3,000,000 = 600,000$
 at 30%: $0.30 \times 2,500,000 = 750,000$ to $0.30 \times 3,000,000 = 900,000$
 at 40%: $0.40 \times 2,500,000 = 1,000,000$ to $0.40 \times 3,000,000 = 1,200,000$

Step 5, the saving as a share of the visit, divided by both performance-restoration ranges as reconciliation rule E.1.4 requires.

blade set only, \$1.6M base, $d = 20\text{--}40\%$: saving 320,000–640,000
 $\div 1,200,000\text{--}1,600,000 = 20\%$ to 53% of the visit; $\div 1,800,000\text{--}2,500,000 = 13\%$ to 36%
 blades and vanes, \$2.5–3.0M, $d = 20\text{--}40\%$: saving 500,000–1,200,000
 $\div 1,200,000\text{--}1,600,000 = 31\%$ to 100% of the visit; $\div 1,800,000\text{--}2,500,000 = 20\%$ to 67%

So on the per-blade inputs, alternative hot-section parts at a 20–40% discount would save 13% to 53% of a restoration's cost on the blades alone and 20% to 100% on blades and vanes together, depending on which visit range is used. A saving approaching or exceeding the whole visit is itself a sign that the per-blade inputs and the visit totals are not on one basis, which is the tension of step 2 seen from the other side. The savings apply only at visits where the blades and vanes are scrapped rather than repaired, which the interview places at second and later visits; at a first visit the lever is a repair (§40). Limits of the example: the blade count, the per-blade price, the vane-set price and the discount are respectively unverified, interview-sourced, assumed within a phrase and assumed; the visit totals are two contested consultancy ranges; and no manufacturer's catalogue price for a blade or a set was read (D4 U16). [D4 §2.8; Safe Fly Aviation, 2026; In Practise, undated; reconciliation R-013, R-015; E. 1 rule 4]

44.3 TESTING THE CLAIMED \$2 MILLION AGAINST THE ARITHMETIC

WORKED EXAMPLE 6.7 — WHAT "UP TO \$2 MILLION PER SHOP VISIT" REQUIRES OF THE FIVE PARTS

Inputs: the claim, "up to \$2 million per shop visit" from five CFM56 hot-section parts supplied to FTAI "at cost" (sourced: investor blogs; not in FTAI filings retrieved). The manufacturer's prices from worked example 6.6: blade set \$1.6M or \$3.08M; blades and vanes \$2.5–3.0M (assumed within the sourced phrase). The other three parts (a low-pressure turbine vane, a shroud by one reading, and a fifth part the commentary does not name): no price retrieved.

Step 1, the effective discount the claim implies if the saving came from the named parts alone.

blade set at \$1.6M:	$2,000,000 \div 1,600,000 = 125\%$	(not reachable from the blade set alone)
blade set at \$3.08M:	$2,000,000 \div 3,080,000 = 65\%$	
blades and vanes at \$2.5M:	$2,000,000 \div 2,500,000 = 80\%$	
blades and vanes at \$3.0M:	$2,000,000 \div 3,000,000 = 67\%$	

Step 2, what is left for the other parts at a market-style discount. At $d = 40\%$ the blade set saves \$640,000 (\$1.6M base), leaving \$1,360,000 to come from the other four parts; at the same discount blades and vanes together save \$1.0–1.2M, leaving \$0.8–1.0M for the low-pressure turbine vane, the shroud and the fifth part, whose prices the research did not reach.

Step 3, the "at cost" reading. If "at cost" means FTAI pays roughly the manufacturing cost rather than a market PMA price, FTAI's saving per part is the manufacturer's list price less the maker's cost, which is larger than the discount a third-party buyer of the same part would receive. Whether that difference is large enough to reach the effective discounts in step 1 depends on the maker's cost, which no source states; the primer shows the required effective discount and does not assume a cost.

Step 4, the claim against the visit totals and the only filed-level comparators.

$2,000,000 \div 1,200,000 - 1,600,000 = 125\% \text{ to } 167\%$ of a restoration on the lower range
 $2,000,000 \div 1,800,000 - 2,500,000 = 80\% \text{ to } 111\%$ of a restoration on the upper range
 $2,000,000 \div 75,000 = 27$ times FTAI's quantified Bristol saving per shop visit
 Southwest, fleet-wide: $54,000,000 \div 341 = 158,358$ per part number per year (not convertible to a per-visit figure)

So the claim is reachable from the blade set alone only at the higher per-blade price and a 65% effective discount, is not reachable from the blade set alone at the interview price, and on either visit range is of the same size as, or larger than, the whole restoration as the consultancy prices it. The arithmetic shows that the claim is sensitive to which per-blade price is right (R-015), to what "at cost" means (R-026), to what the other three parts cost (unknown), and to what "shop visit" the blogs mean, since a heavy visit is larger than a restoration (R-013). The example neither confirms nor refutes the claim; it sets out what would have to be true. Limits of the example: every price input is third-party, unverified or assumed; the claim's own basis (against list price, at which visit, "up to") is not stated by its sources; and FTAI has published no per-visit PMA saving. [Komodo Capital; Kairos Research; D4 §2.8 step 4, T1, T5; Insider Monkey, Oct 2025; Aviation Week, c. 2023; reconciliation R-013, R-015, R-026, R-027]

WHAT THE PMA LEVER DOES NOT TOUCH

Whatever the right per-blade price, an approved alternative part changes only the on-condition parts line of a shop visit. Life-limited parts, 40–60% of parts cost when they are replaced, remain single-source at the manufacturer's escalating list price, because no PMA exists for any CFM56 disk, spool or shaft. Labour, test and the shop's margin are untouched. And at a visit where the blades can still be repaired, no new blade is bought from anyone, so the lever is a repair, not a part. The reach of the PMA lever is therefore the new hot-section parts bought at second and later visits, which is a real and growing share of visits on a fleet kept flying past its planned retirement (Part III §19), and no more than that. [Safe Fly Aviation, 2026; D4 U7, §2.7; In Practise, undated]

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-013	Performance restoration \$1.2–1.6M and \$1.8–2.5M; heavy with LLPs \$2.1–2.8M and ">\$3.5M"; full overhaul \$3.5–4.2M (all Safe Fly Aviation, consultancy blog, 2026); PMA savings divided by both PR ranges	\$40.3 worked example 6.2; \$44 worked examples 6.6 and 6.7

Id	The two (or more) figures	Where in this Part
R-014	Material 60–70% / labour 20–30% / repairs 10–20% (Air Cargo Week) against LLPs 40–60% of parts / HPT blades and vanes 15–25% / labour 15–20% (Safe Fly)	§37.1; §40.3; §44.1
R-015	HPT blade set \$1.6M (80 × \$20,000, interview) or \$3.08M (80 × ~\$38,500, weak listing) against \$0.11–0.63M implied by the 15–25% share on either visit range	§37.1 worked example 6.1; §44.2 worked example 6.6 step 2; §44.3
R-018	Escalation: spares ~10%/yr, double-digit Nov 2022, high-single-digit Aug 2023, inflation +3–4 points (Aviation Week) against LLP set ~7%/yr compounded (derived) and -7B parts +20–30% in 2023 (Magnetic via Aviation Week)	§37.1
R-023	PMA market: \$11.8B in 2025 (IMARC) against \$14B by 2026 (GIA, 2022); engine PMA+DER \$2.95B in 2022 (GII listing) against "engines the largest segment" of \$11.8B; FlightGlobal ceiling ~21% against a doubling forecast	§41.4, Exhibit 6.7
R-024	HEICO new PMAs per year: "approximately 300 to 500" (D4's reading) against "approximately 400 to 550" (D11's reading), both HEICO 10-K FY2025	§38.3; §41.1, Exhibit 6.5
R-025	Chromalloy CFM56-5B/7B roster: three announced approvals (blade, HPT vane 1, LPT vane; D4) against four parts offered including the HPT shroud (D5's reading of the Newswire copy); blogs' "two approved" and "second of five"	§41.2, Exhibit 6.6; §43.2
R-026	The relationship: "a joint venture" (FTAI 10-K FY2025) against "exclusive perpetual partnership", parts "at cost", formed 2018 (investor blogs) against a 25% "Advanced Engine Repair JV" first funded Dec 2016, identity unknown (orchestrator reading of the 10-K)	§43.1, Exhibits 6.11 and 6.12
R-027	Saving: "up to \$2 million per shop visit" and "5–10 percentage points of margin" (investor blogs) against FTAI's qualitative "cost savings and flexibility" and its one quantified lever, \$75,000 per shop visit (Bristol JV, Q3 2025 call)	§43.4 worked example 6.5; §44.3 worked example 6.7
D4 T6	"Most lessors permit PMA ... with a few exceptions" (MARPA, Nov 2015, aircraft leases) against engine lessors forbidding PMA and DER "in the vast majority of cases" (ELFC, mid-2016)	§42.2, Exhibit 6.9
D4 T7	PMA share of engine parts "capped at around 21%" (FlightGlobal) against the engine PMA+DER market doubling 2022–2032 (vendor forecast)	§41.4

Id	The two (or more) figures	Where in this Part
D4 T8	CFM's 2018 Conduct Policies (no warranty voiding, no service refusal) alongside TRUEngine and residual-value messaging for OEM-only configuration	§42.3

Unknowns carried in this Part

- **U-D4-01 (D4): U1 — Current lessor return-condition practice (2024–2026).** All lessor statements found are 2009–2016. Resolve with recent lessor 10-K/20-F lease descriptions (AerCap, Air Lease, Willis Lease), ISTAT or IBA surveys, or lessor conference remarks.
- **U-D4-02 (D4): U2 — PMA price discount for the Chromalloy HPT blade and vane** (and HEICO's typical discount). Resolve with Chromalloy/FTAI pricing disclosure or MRO price catalogues (the SR Technics 2026 CFM56-7B catalogue may list PMA options [71]).
- **U-D4-03 (D4): U3 — FTAI–Chromalloy JV terms:** formation date, ownership split, accounting (consolidated vs equity method), exclusivity, "at cost" pricing, revenue recognized. Resolve by reading FTAI's FY2018–FY2025 10-Ks for the JV footnote and any 8-K. (Partly and conditionally closed by the orchestrator notes: Advanced Engine Repair JV 25%, \$15.0M Dec 2016 + \$13.5M Aug 2019, carrying value \$22,429K at 31 Dec 2025, if the same entity; reconciliation D.3, K-04.)
- **U-D4-04 (D4): U4 — Time from PMA application to approval** for hot-section parts. Resolve via FAA PMA database (DRS) entries with dates, or company statements.
- **U-D4-05 (D4): U5 — Whether the CFM56 HPT blade/vane PMAs are "critical components"** under the TIP and therefore which EASA acceptance clause applies; whether EASA acceptance has been obtained.
- **U-D4-06 (D4): U6 — Acceptance by other regulators** (UK CAA, TCCA, ANAC, CAAC, DGCA India) of these PMAs.
- **U-D4-07 (D4): U7 — Any PMA for CFM56 life-limited rotating parts.** None found.
- **U-D4-08 (D4): U8 — How DER-approved repair data is accepted on EU-registered engines** under the TIP.
- **U-D4-09 (D4): U9 — HEICO portfolio split** by part category (consumables vs rotating vs hot-section) and HEICO's PMA discount.
- **U-D4-10 (D4): U10 — Any CFM/GE statement specifically on the Chromalloy HPT blade PMA,** and any GE "material" programme (e.g. TrueChoice Material) positioning against PMA. The brief's phrase "material solutions" was not matched to a current CFM/

GE programme in this session; the historical P&W "Global Material Solutions" CFM56-3 PMA venture was found only via headlines [51][52].

- **U-D4-11 (D4): U11 — PMA policies of Delta, American, United, Ryanair, and non-US CFM56 operators** for hot-section parts.
- **U-D4-12 (D4): U12 — Chromalloy production capacity** (blade sets per year) and FTAI's share of it.
- **U-D4-13 (D4): U13 — Quantified residual-value effect** of PMA/DER content on engine or aircraft appraisals.
- **U-D4-14 (D4): U14 — Terms of the FTAI–CFM Jan 2026 materials agreement** (duration, volumes, pricing, whether it restricts FTAI's PMA use).
- **U-D4-15 (D4): U15 — Customer uptake figures for the Chromalloy blade** (named airlines/MROs, sets delivered). Only "sold out for 2025" was found.
- **U-D4-16 (D4): U16 — CFM56-7B HPT blade count per set** (80 is a recollection, unverified here) and the OEM list price for a full HPT blade and vane set.
- **U-D4-17 (D4): U17 — Dates for the LPT vane PMA and the V2500 HPT blade PMA.**
- **U-D5-11 (D5): U11 — Chromalloy JV contribution in FTAI's numbers:** equity-method earnings line, PMA share of parts, FTAI-stated savings per module. (Cluster K-04 with U-D4-03.)
- **U-D11-05 (D11): U5 — HEICO consolidated FY2025 net sales and operating income; PMA cumulative count; stated discount to OEM; acceptance statements.** Searched [S7, S7a, S8]. Only FSG figures and the 400–550 PMA rate were returned. Resolve: read Item 1 and Item 7 of [S7]. (Cluster K-05 with U-D4-02 and U-D4-09.)
- **U-D10-11 (D10): U-11 — Early asset mix and the shift to aviation.** Figures for Jefferson Terminal, Repauno, Transtar and Long Ridge; dates and prices of QuickTurn, the Chromalloy JV and the Lockheed Martin Commercial Engine Solutions acquisition. Resolve with the FY2021 10-K [S15], the FY2022 annual report and the relevant 8-Ks. (Carried for the Chromalloy JV deal terms only; cluster K-12; the facility items are partly closed and are Part VII's and Part X's.)

Sources

Regulatory 1. eCFR, 14 CFR Part 21 Subpart K — Parts Manufacturer Approvals (current). <https://www.ecfr.gov/current/title-14/chapter-I/subchapter-C/part-21/subpart-K> 2. FAA, Advisory Circular 21.303-4 (PMA application). https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_21_303-4.pdf 3. FAA, AC 21.303-3, Application for PMA via Tests and Computations or Identicality. https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/1023551 4. FAA, Order 8110.42D, Parts

Manufacturer Approval Procedures (with change 1). https://www.faa.gov/documentLibrary/media/Order/FAA_Order_8110.42D_w-chg_1.pdf 5. FAA, Parts Manufacturer Approval — Design Approval (web page). https://www.faa.gov/aircraft/air_cert/design_approvals/pma/pma_des 6. FAA, AC 33-8, Guidance for PMA of Turbine Engine and APU Parts under Test and Computation. https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/99620 7. FAA, Notice N 8110.55, PMA by Identity, 19 July 1996. <https://www.faa.gov/documentLibrary/media/Order/N8110-55.pdf> 8. FAA–EASA, Technical Implementation Procedures Rev 7 with Amendment 1. https://www.faa.gov/aircraft/air_cert/international/bilateral_agreements/eu/tip/tip_rev7_with_amdt1_incorporated 9. FAA–EASA, Technical Implementation Procedures Rev 6 with Amendments 1 and 2 (PDF). https://www.faa.gov/sites/faa.gov/files/EUTIP_Rev6_w_amdt1_amdt2.pdf 10. FAA, EASA Frequently Asked Questions (PDF). https://www.faa.gov/sites/faa.gov/files/aircraft/air_cert/international/easa/EASA_FAQ.pdf 11. FAA, Order 8110.37F, Designated Engineering Representative (DER) Handbook. https://www.faa.gov/documentLibrary/media/Order/FAA_Order_8110.37F.pdf 12. FAA, Documenting Compliance Findings (Form 8110-3 guidance). https://www.faa.gov/sites/faa.gov/files/other_visit/aviation_industry/designees_delegations/individual_designees/compliance_findings_8110-3.pdf 13. FAA, Order 8300.16, Major Repair and Alteration Data Approval. https://www.faa.gov/documentLibrary/media/Order/8300_16.pdf 14. eCFR / FAA, 14 CFR 45.15 (marking of PMA articles), referenced through source 1 and source 5.

HEICO (sec.gov) 15. HEICO Corp, Form 10-K for fiscal year ended 31 Oct 2025. <https://www.sec.gov/Archives/edgar/data/46619/000004661925000082/hei-20251031.htm> 16. HEICO Corp, Q4/FY2025 earnings release (Ex. 99.1 to 8-K), 18 Dec 2025. <https://www.sec.gov/Archives/edgar/data/46619/000004661925000076/a10312025ex991earningsrele.htm> 17. HEICO Corp, Q1 FY2025 earnings release (Ex. 99.1), Feb 2025. <https://www.sec.gov/Archives/edgar/data/46619/000004661925000013/a01312025q1ex991earningsre.htm> 18. HEICO Corp, press release on acquisition of Wencor Group (Ex. 99.1), 2023. https://www.sec.gov/Archives/edgar/data/46619/000095014223001445/eh230358705_ex9901.htm

FTAI (sec.gov, GlobeNewswire) and orchestrator notes 19. FTAI Aviation Ltd., Form 10-K for FY2025 (filed early 2026). <https://www.sec.gov/Archives/edgar/data/1590364/000162828026012940/ftai-20251231.htm> 20. FTAI Aviation Ltd., Form 10-Q for quarter ended 30 June 2026. <https://www.sec.gov/Archives/edgar/data/0001590364/000162828026051412/ftai-20260630.htm> 21. FTAI Aviation, "FTAI Aviation Announces Multi-Year Materials Agreement with CFM International to Further Support CFM56 Engines," GlobeNewswire, 22 Jan 2026. <https://www.globenewswire.com/news-release/2026/01/22/3223741/35538/en/FTAI-Aviation-Announces-Multi-Year-Materials-Agreement-with-CFM-International-to-Further-Support-CFM56-Engines.html> 22. FTAI Aviation Ltd., Q4/FY2025 earnings release (8-K Ex. 99.1), 25 Feb 2026. <https://www.sec.gov/Archives/edgar/data/>

1590364/000162828026011685/ftai123125earningsrelease.htm 23. FTAI Aviation Ltd., QuickTurn Europe joint venture closing, GlobeNewswire, 5 Jun 2025. <https://www.globenewswire.com/news-release/2025/06/05/3094782/0/en/ftai-aviation-ltd-announces-closing-of-quickturn-europe-joint-venture.html> 24. Orchestrator notes, "Equity-method investments (10-K FY2025; 10-Q Q3 2025 note R13)", /home/user/SPOT/ftai-primer/research/orchestrator-notes.md, 2026-10-03 (reading of source 19 and the Q3 2025 10-Q). 25. MarketBeat, "FTAI Aviation Q4 earnings call highlights," 26 Feb 2026 (call coverage). <https://www.marketbeat.com/instant-alerts/ftai-aviation-q4-earnings-call-highlights-2026-02-26/> 26. Motley Fool, FTAI Q4 2025 earnings call transcript, 26 Feb 2026 (call coverage). <https://www.fool.com/earnings/call-transcripts/2026/02/26/ftai-ftai-q4-2025-earnings-call-transcript/> 27. Insider Monkey, FTAI Q3 2025 earnings call transcript, Oct 2025 (call coverage). <https://www.insidermonkey.com/blog/ftai-aviation-ltd-nasdaqftai-q3-2025-earnings-call-transcript-1635931/>

Chromalloy and the PMA roster 28. Chromalloy, "Chromalloy Secures FAA Approval of CFM56 High Pressure Turbine Blade PMA," 30 Oct 2025. <https://www.chromalloy.com/chromalloy-secures-faa-approval-of-cfm56-high-pressure-turbine-blade-pma/> 29. Newswire.com, same release, 30 Oct 2025. <https://www.newswire.com/news/chromalloy-secures-faa-approval-of-cfm56-high-pressure-turbine-blade-22665618> 30. AviTrader, "Chromalloy wins FAA approval for CFM56 HPT blade," 31 Oct 2025. <https://avitrader.com/2025/10/31/chromalloy-wins-faa-approval-for-cfm56-hpt-blade/> 31. Aviation Maintenance Magazine, "Chromalloy Secures FAA Approval of CFM56 High Pressure Turbine Blade PMA," 2025. <https://avm-mag.com/chromalloy-secures-faa-approval-of-cfm56-high-pressure-turbine-blade-pma> 32. Chromalloy, "Chromalloy Receives FAA Approval of CFM56-5B/7B HPT Turbine Vane 1 PMAs," 2025. <https://www.chromalloy.com/chromalloy-receives-faa-approval-of-cfm56-5b-7b-hpt-turbine-vane-1-pmas/> 33. Aeromorning, "Chromalloy Secures FAA Approval of CFM56 LPT Vane Up-Change PMA" (undated). <https://aeromorning.com/en/chromalloy-secures-faa-approval-of-cfm56-lpt-vane/> 34. Aviation Week, "Chromalloy Unveils FAA-Approved V2500 HPT Blade," MRO Europe (date not captured). <https://aviationweek.com/shows-events/mro-europe/chromalloy-unveils-faa-approved-v2500-hpt-blade> 35. PRNewswire, "U.S. Air Force Selects Chromalloy Joint Venture Company — BELAC LLC — to Provide High Pressure Turbine Blades for F108 Aircraft Engines" (2000s). <https://www.prnewswire.com/news-releases/us-air-force-selects-chromalloy-joint-venture-company---belac-llc---to-provide-high-pressure-turbine-blades-for-f108-aircraft-engines-103934938.html> 36. GovConWire, "Veritas Buys Sequa From Carlyle, Adds Chromalloy to Portfolio," Dec 2022. <https://govconwire.com/2022/12/veritas-buys-sequa-from-carlyle-adds-chromalloy-to-portfolio> 37. Bloomberg, "Carlyle Is Said to Explore \$2 Billion Sale of Sequa Business," 19 July 2022. <https://www.bloomberg.com/news/articles/2022-07-19/carlyle-is-said-to-explore-2-billion-sale-of-sequa-business> 38. FlightGlobal,

"The sum of the parts" (Chromalloy profile), undated. <https://www.flightglobal.com/the-sum-of-the-parts/96138.article> 39. PRNewswire, "AAR signs exclusive PW4000 agreement with Chromalloy," Mar 2025. <https://www.prnewswire.com/news-releases/aar-signs-exclusive-pw4000-agreement-with-chromalloy-302392076.html> 40. PRNewswire, "AAR signs new engine parts supply agreement with Chromalloy," Nov 2024. <https://www.prnewswire.com/news-releases/aar-signs-new-engine-parts-supply-agreement-with-chromalloy-302301855.html>

CFM / GE / IATA 41. CFM International, "CFM boosts HPT blade output and repair to meet extended demand for CFM56 engines" (2025–26; date not captured). <https://www.cfmaeroengines.com/stories/cfm-boosts-hpt-blade-output-and-repair-to-meet-extended-demand-for-cfm56-engines> 42. CFM International, "Investing in Durability: CFM ships 1,000th set of new CFM56 HPT blades." <https://www.cfmaeroengines.com/stories/investing-in-durability-cfm-ships-1000th-set-of-new-cfm56-hpt-blades> 43. Safran / CFM, "CFM introduces upgraded HPT blade for CFM56 engines," 16 Oct 2023. <https://www.safran-group.com/pressroom/cfm-introduces-upgraded-hpt-blade-cfm56-engines-2023-10-16> 44. IATA, "IATA–CFM Agreement on Engine Maintenance" (2018, PDF). <https://www.iata.org/contentassets/b7fc716af6a94192b1889420c7d573ce/iata-cfm-agreement-on-engine-maintenance.pdf> 45. IATA, "CFM Conduct Policies and Implementation Measures" (PDF). <https://www.iata.org/contentassets/f03b1a4b79534b99802f10cd23b19ec2/cfm-conduct-policies-and-implementation-measures.pdf> 46. CFM International, "CFM External Guidance" (Mar 2019, PDF). https://www.cfmaeroengines.com/wp-content/uploads/2019/03/CFM_External_Guidance.pdf 47. MARPA, "CFM agrees to permit the use of PMA parts and DER repairs," 1 Aug 2018. <https://marpa.pmaparts.org/2018/08/01/cfm-agrees-to-permit-the-use-of-pma-parts-and-der-repairs/> 48. Aviation Suppliers Association, Member Bulletin Oct 2018, "CFM Agrees to permit the use of PMA Parts and DER Repairs." <https://www.aviationsuppliers.org/ASA-Member-Bulletin---Oct-2018---CFM-Agrees-to-permit-the-use-of-PMA-Parts-and-DER-Repairs> 49. GE Aerospace, TrueEngine presentation (PDF). <https://www.geaerospace.com/sites/default/files/trueengine-presentation.pdf> 50. GE Aerospace / CFM, "CFM expands TRUEngine program; unique content assurance guarantee [for] subsequent [buyers]." <https://geaerospace.com/news/press-releases/services/cfm-expands-truengine-program-unique-content-assurance-guarantee-subsequent> 51. Aviation International News, "CFM TRUEngine program to cover subsequent buyers," 12 Sept 2012. <https://backend.ainonline.com/aviation-news/business-aviation/2012-09-12/cfm-truengine-program-cover-subsequent-buyers> 52. FlightGlobal, "Farnborough: P&W parts display fuels CFM spat" (Guy Norris), undated. <https://www.flightglobal.com/farnborough-pandw-parts-display-fuels-cfm-spat/68616.article> 53. FlightGlobal, "P&W to conclude PMA parts tests in six weeks," undated. <https://www.flightglobal.com/pandw-to-conclude-pma-parts-tests-in-six-weeks/77119.article> 54. General Electric Co. (GE Aerospace), Form 10-K FY2025 (filed Feb 2026). <https://>

www.sec.gov/Archives/edgar/data/40545/000004054526000008/ge-20251231.htm 55. Aviation Week, "Parts Price Hikes Help Boost Safran, GE Aftermarket Sales" (late 2023). <https://ngtest.aviationweek.com/mro/aircraft-propulsion/parts-price-hikes-help-boost-safran-ge-aftermarket-sales> 56. Aviation Week, "CFM Aftermarket Shifting To Long-term Agreements, Safran Says" (c. 2023/24). <https://aviationweek.com/mro/aircraft-propulsion/cfm-aftermarket-shifting-long-term-agreements-safran-says>

Lessors and airlines 57. MARPA, "Are leases getting looser?", 10 Nov 2015. <https://marpa.pmaparts.org/2015/11/10/are-leases-getting-looser/> 58. Engine Lease Finance Corporation (ELFC), "An overview of the aircraft engine aftermarket in mid 2016." <https://elfc.com/an-overview-of-the-aircraft-engine-aftermarket-in-mid-2016/> 59. Aviation Suppliers Association, "Airlines: now is the time to choose lessors who accommodate PMA and DER repairs," undated. <https://www.aviationsuppliers.org/airlines-now-is-the-time-to-chose-lessors-who-accommodate-pma-and-der-repairs> 60. Aircraft Commerce, Issue 67 (2009), maintenance article on PMA and lessors (PDF). https://aircraft-commerce.com/wp-content/uploads/aircraft-commerce-docs/Maintenance/2009/ISSUE67_MTCE_A.pdf 61. Sofema Online, "Aircraft Lease Management — Continuing Airworthiness In-Service Considerations," undated. <https://legacy.sofemaonline.com/component/easyblog/aircraft-lease-management-continuing-airworthiness-in-service-considerations?Itemid=101> 62. Aviation Week, "Southwest's 737-700 Engine Strategy As It Phases Out Fleet" (c. 2023). <https://ngtest.aviationweek.com/mro/aircraft-propulsion/southwests-737-700-engine-strategy-it-phases-out-fleet> 63. Fox Business, "Major US airlines find unapproved jet engine parts in some aircraft as alleged supplier faces lawsuit," 2023. <https://www.foxbusiness.com/lifestyle/major-us-airlines-find-unapproved-jet-engine-parts-some-aircraft-alleged-supplier-faces-lawsuit> 64. Aircraft Value News, "PMA continues to undermine full-life maintenance adjustments," undated. <https://www.aircraftvaluenews.com/pma-continues-to-undermine-full-life-maintenance-adjustments/>

Market size (vendor and trade) 65. IMARC Group, "Commercial Aircraft Parts Manufacturer Approval Market" (2025 base). <https://www.imarcgroup.com/report/en/commercial-aircraft-parts-manufacturer-approval-market> 66. GII Korea listing, "Global Engine PMA Parts & DER Repair Market Research" (2022 base). <https://www.giikorea.co.kr/report/marf1540758-global-engine-pma-parts-der-repair-market-research.html> 67. PRNewswire / Global Industry Analysts, "...Commercial Aircraft PMA with the Market to Reach \$14 Billion Worldwide by 2026," 2022. <https://prnewswire.com/news-releases/new-analysis-from-global-industry-analysts-reveals-steady-growth-for-commercial-aircraft-pma-with-the-market-to-reach-14-billion-worldwide-by-2026-301488219.html> 68. Persistence Market Research, "Aircraft Aftermarket Parts Market" (2026–2033). <https://www.persistencemarketresearch.com/market-research/aircraft-aftermarket-parts-market.asp> 69. FlightGlobal, "Aftermarket shocks" (PMA share capped ~21%; undated). <https://flightglobal.com/aftermarket-shocks/73224.article>

Shop-visit cost inputs (via Parts I and IV) 70. Air Cargo Week, "The true cost of engine maintenance" (2025/26). <https://aircargoweek.com/the-true-cost-of-engine-maintenance/> 71. Safe Fly Aviation (consultancy blog, 2026): "Engine Shop Visit Costs Worldwide 2026" <https://safefly.aero/?p=15606> ; "What determines aircraft engine overhaul costs?" <https://safefly.aero/blog-aircraft-engine-overhaul-cost-drivers/> ; "CFM56-7B Engine Availability Report 2026" <https://safefly.aero/cfm56-7b-engine-availability-report/> 72. Aviation Week, "Magnetic expects CFM56 market challenges" (-7B parts +20–30% in 2023), 2024, via Part I §4.3 and D1.

Third-party commentary on FTAI–Chromalloy (not FTAI disclosure) 73. In Practise, "FTAI, Chromalloy and CFM56 HPT Blade PMA" (expert interview), undated. <https://inpractise.com/articles/ftai-chromalloy-and-cfm56-hpt-blade-pma> 74. In Practise, "FTAI & PMA Parts, ..." (interview digest), undated. <https://inpractise.com/articles/ftai-and-pma-parts-hermes-rolex-veralto-ccc-avon-cvna-lgi-w-didi> 75. Komodo Capital (Substack), "Why FTAI Aviation (\$FTAI) Presents a Rare Opportunity," undated. <https://komodocapital.substack.com/p/why-ftai-aviation-ftai-will-fly-in> 76. Kairos Research (Substack), "Why I bought FTAI Aviation," undated. <https://kairosresearch.substack.com/p/why-i-bought-ftai-aviation> 77. Yahoo Finance, "FTAI Aviation Ltd. (FTAI): A Bull Case Theory," undated. <https://finance.yahoo.com/news/ftai-aviation-ltd-ftai-bull-235824752.html>

Price inputs 78. SR Technics (SR), "CFM56-7B Capabilities & Price Catalogue," 3 June 2026 (PDF; located, not read). <https://www.srtechnics.com/media/zgcnmf3o/cfm56-7b-sr-technics-capabilitiesprice-catalogue-03062026.pdf> 79. Web page on a university web host (ATU) listing a CFM56-7B HPT blade at about €35,000 (provenance weak; condition unclear), undated. <https://g00431067.webhosting.atu.ie/?p=1313>

Primer cross-references 80. Part I of this primer, "First principles: the CFM56 as a machine", §4.3 (LLP set price series and derived escalation), §5, §7.2 (Exhibit 1.10) and §9; Part IV §28–29; Part V §31, §34; Part VII §52; reconciliation.md sections A.2, A.4, C.4, D and E; term-registry.md Part VI rows and sections 1b, 1c, 2 and 3.